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MILITARY SPECIFICATIONS

COMBAT CHARTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines requirements for the Defense Mapping Agency's (DMA) Combat Charts.

1.2 Purpose. The purpose of this specification is to assure uniformity of treatment among all mapping and charting elements engaged in a coordinated production program for this product.

1.3 Security.

1.3.1 Security Classification. The security classification of the products generated by the use of these specifications will be the lowest category practicable. When it is necessary to assign a security classification to the product, it shall be in accordance with established national security procedures.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, Defense Mapping Agency, ATTN: PR, 8613 Lee Highway, Fairfax, VA 22031-2137 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **AREA MCGT**

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the current Department of Defense Index of Specifications and Standards (DODISS) and the supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MIL-T-89301

STANDARDS

MILITARY

MIL-STD-600000	-	MC&G glossary of Feature/ Attribute Definitions.
MIL-STD-600001	-	MC&G Accuracy.
MIL-STD-600002	-	MC&G Symbology.
MIL-STD-600003	-	MC&G Product Generation Rules.
MIL-STD-600004	-	MC&G Geographic Names.
MIL-STD-600005	-	MC&G Reproduction and Printing.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Chart No. 1	-	Nautical Chart Symbols and Abbreviations
PUBS 110-116 (LLPUB)	-	List of Lights
NM	-	Notice to Mariners
PUB117	-	Radio Navigation Aids
SDPUB	-	Sailing Directions
DMA TM 8358.1	-	Datums, Ellipsoids, Grids and Grid Reference Systems
STANDCONTABLE 04	-	Standard Conversion Table No.4

(Copies of the above publications are available from the Defense Mapping Agency Combat Support Center, Washington, DC 30215-0010).

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2.2 Non-Government publications.

MP004 - Chart Specifications of the IHO and Regulations of the IHO for International (INT) Charts

(Copies of the above publication are available from the International Hydrographic Bureau - Monaco)

NP139 - Echo Sounding Correction Tables (3rd or latest edition)

(Copies of the above publication are available from the British Admiralty, Taunton, U.K.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards) the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Accuracy.

3.1.1 Accuracy requirement (topographic features).

Intended Use	Absolute Horizontal (90% CE to Preferred Datum)	Absolute Vertical (90% LE to MSL)
Amphibious Warfare	50 meters	2 contour intervals
Gunfire Support	50 meters	1 contour interval
Antisubmarine Warfare	100 meters	N/A
Mine Warfare	100 meters	N/A
Training	50 meters	1 contour interval
Evacuation	50 meters	1 contour interval

3.1.2 Displaced features. Feature symbols which are displaced, as identified in Section 3.24 of this specification (reference: displacement rules found in MIL-STD-600003) are excluded from the accuracy requirement stated above.

3.1.3 Hydrographic information accuracy. The hydrographic area of Combat Charts shall be compiled with the most accurate depth information available. The charts are compiled from a variety of sources with varying accuracies. The most accurate data is that taken from hydrographic surveys which meet IHO survey accuracy standards. IHO survey accuracy standards are as follows.

3.1.3.1 Vertical accuracy.

± 0.3 meters at 90% LE (see 6.3.2) for depths <= 30 meters.
± 1.0 meters at 90% LE for depths > 30 meters and <= 100 meters.
± 1% of depth at 90% LE for depths > 100 meters.

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3.1.3.2 Horizontal accuracy.

50 meters at 90% CE (see 6.3.1) relative to World Geodetic System 1984 (WGS 84).

3.2 Datum.

3.2.1 Horizontal datum. For new production, and as map/chart sheets are revised or updated for periodic maintenance, the WGS 84 or NAD 83 datum and where appropriate a revised Military Grid system shall be depicted as the primary grid. The old (local) datum will be depicted as a secondary grid with tick marks along the border of the sheet. A grid conversion note shall also be placed in the margin area. Additionally, both the old and new 100,000 meter square two-letter identifiers shall be depicted on the map/chart, if applicable. Appropriate margin notes shall be added to explain the dual lettering.

3.2.2 Vertical datum. Combat Charts shall have three vertical datums. The topographic features are referenced to Mean Sea Level. The shoreline (2A010) is referenced to Mean High Water. Hydrographic features are referenced to a vertical datum based on a low water tide level and called the sounding datum or hydrographic datum. The specific low water datum used depends on the type of tide in the area or the number and magnitude of high and low tides in one tidal cycle.

3.3 Adjoining data set and chart match.

3.3.1 Adjacent Combat Charts. Adjacent Combat Charts shall have an area of overlap between them to make it easier to transfer positions from one chart to the next.

3.3.2 Chart detail. When adjoining charts are compiled at the same time, chart detail in the overlap area must be identical, except in cases where the newest Combat Charts are based on later date sources. In these cases, the new Combat Charts shall show the later information. If the differences in the overlap area are critical for safe navigation, the previously published Combat Charts shall be updated by Classified Notices to Mariners.

3.3.3 New or new edition. When new or new edition Combat charts are produced and the adjacent charts are not updated, chart detail in the overlap area shall be identical to the previously published charts, except as noted in 3.3.2.

3.4 Series. Combat Charts produced as part of a series shall be outlined in a location diagram in the margin.

3.5 Quality. Final product quality shall reflect the quality elements expressed by each appropriate section within MIL-STD-600000, MIL-STD-600001, MIL-STD-600002, MIL-STD-600003, MIL-STD-600004 and MIL-STD-600005.

3.6 Scale. Combat Charts and Combat Training Charts are constructed at a standard scale of 1:50,000.

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3.7 Chart design.

a. Combat Charts provide chart coverage for approximately fifteen nautical miles landward and fifteen nautical miles seaward (total thirty nautical miles); however, the proportion of land and water may vary for special user requirements. The charts shall be oriented East-West or North-South. The long edge of a Combat Chart runs generally perpendicular to the coastline to provide sufficient coverage inland and out to sea.

b. Combat Charts are rectangular in shape, except where a small island falls just outside the limits of a chart, and is not shown on another chart in the series, an extension is added to the chart to include the island, rather than make a new chart just for the island.

c. An area of overlap is provided for ease of transition from one chart to the next. The width of this overlap area is variable but shall not be less than 1 minute or greater than 2 minutes.

3.8 Size and dimensions. The standard trim size for charts oriented East-West or North-South is 105.4 cm by 147.3 cm (41-1/2 inches by 58 inches). The standard neatline size for East-West chart is 78.7 cm by 137.2 cm(31 inches by 54 inches). The standard neatline size for North-South charts is 81.3 cm by 134.6 cm (32 inches by 53 inches). See Appendixes A and B (Style Sheets).

3.9 Projection. Combat Charts are constructed on the Transverse Mercator projection.

3.9.1 Graticule lines and subdivisions.

a. Parallels and meridians are shown in black (SPC-58600) at 4 minute intervals, except as noted in 3.9.4.

b. Full minute ticks are shown along the neatline.

c. Minute ticks are shown along a selected central meridian and a selected central parallel. The central meridian and parallel which is subdivided is selected on the basis of clarity of presentation with preference given to open water area wherever possible. Another major consideration in the portrayal of subdivisions along projection lines is the location of rows of grid labels. See Appendixes A and B for examples of minute tick subdivisions.

d. Five second ticks subdividing one minute are shown at two selected locations on each neatline, central meridian and central parallel. Locations are selected on the basis of clarity of presentation with preference given to open water areas.

3.9.2 Graticule labeling.

a. Geographic coordinates showing degrees, minutes, and seconds are shown on all four neatline corners. The appropriate suffix N, S, E, or W is shown with full coordinate values (degrees, minutes, and seconds) at the lower left and upper right corners.

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b. Parallels, meridians and minute tick labels are shown along the neatline. The subdivided central parallel and meridian labels show the degree and minute values at the neatline.

c. Every minute tick on the subdivided central parallel and meridian shows a minute label except at the 4 minute graticule which shows a label with the whole degree and minute value.

d. Five second subdivision ticks are shown at the 15", 30", and 45" ticks.

3.9.3 Additional graticule information. Refer to Appendices A and IB, Style Sheets for additional information about labeling and subdivision.

3.9.4 Exception to graticule intervals. Although meridians and parallels are usually shown at 4 minute intervals, this interval becomes too dense (approximately 3 inches) at 60 degrees latitude and greater. Whenever the graticule interval measures less than 7.6 cm (3 inches), the interval is increased. For Combat Charts at 60 degrees latitude and greater, the interval for meridians is 10 minutes and the interval for parallels is 5 minutes.

3.9.5 Graticule and grid accuracy.

a. The intersections of parallels and meridians shall be within 0.1mm of computed positions.

b. The overall distance between the first full grid lines adjacent to opposite neatlines shall not vary by more than 0.3mm from their computed measurements. The distances between adjacent grid lines will not vary by more than 0.1mm from the computed grid interval.

3.10 Reference systems.

a. The Military Grid system is shown in accordance with the Defense Mapping Agency Technical Manual 8358.1 "Datums, Ellipsoids, Grids, and Grid Reference Systems."

(1) The first major or primary grid is shown in purple (SPC-96532). The secondary grid is shown in blue (SPC-47651). If a third grid is required, it is shown in red-brown (SPC-61121).

(2) The 1000 meter Northing grid lines are labeled to the right of each 10,000 meter Easting grid line and the 1000 meter Easting grid lines are labeled above each 10,000 meter Northing grid line.

b. The dimensions, size and style, and placement of marginal data relating to grids and grid formats are contained in the style sheets (Appendices A and B).

3.11 Margin data.

3.11.1 General. Refer to Appendix A - East-West Style Sheet and Appendix B - North-South Style Sheet for graphic illustrations of the design, the composition, and location of margin data. Notes and diagrams that will vary from chart to chart are indicated by

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accompanying notation on the style sheets. Variations on those examples are explained in this section.

3.11.2 Accuracy diagram. Accuracy Diagrams are shown on all charts. The diagram shows the shoreline, the outlines of individual map sheets used in the compilation, corresponding sheet numbers, outlines of hydrographic data sources, corresponding letter designators keyed to the compilation data list, an outline of areas covered by differing contour intervals if more than one interval is used, and land areas that do not meet the required topographic accuracy. The compilation diagram is shown in black (SPC-58600) except for the map sheet outlines and numbers in green (SPC-52813), the topographic contour interval information in red-brown (SPC-61121), water tint in blue (SPC-47651, 21% screen), and areas that do not meet required accuracy in purple (SPC-96532, LP-15). The diagram is located in the lower chart margin on East-West oriented charts and in the right margin on North-South oriented charts.

3.11.3 Accuracy note.

a. If the accuracy of the chart meets the required topographic accuracy, the following note is shown:

> All Land areas of this chart provide accuracy meeting the criterion that 90% of all well defined features, except those unavoidably displaced by symbol exaggeration, are located within 50 meters of their geographic position with reference to the prescribed datum.

b. If the accuracy of the chart does not meet the required topographic accuracy, the following note is shown:

> Each cross-hatched area in the Accuracy Diagram above does not provide topographic accuracy meeting the criterion that 90% of all well defined features, except those unavoidably displaced by symbol exaggeration, are located within 50 meters of their geographic positions with reference to the prescribed datum.

c. The Accuracy Note is shown in black (SPC-58600) in 6 point Swiss 742, upper and lower case, below the Compilation Data on North-South charts, and below the Accuracy Diagram on East-West charts:

d. This accuracy note does not apply to hydrographic accuracy.

3.11.4 Bar Scales. Bar Scales are shown on each chart. They are shown in black (SPC-58600) and include statute miles, nautical miles and meters. Expanded Bar scales indicating 40,000 yards are shown in the chart margins, parallel to the neatlines on opposite sides of the chart and adjacent to the water area.

3.11.5 Catalog number. Catalog Numbers are no longer shown on Combat Charts as there is now one catalog for all charts.

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3.11.6 Cautions and general notes.

a. Cautions are notes to the mariner warning of specific dangers that exist in specific geographic positions within the chart, or dangerous conditions that apply to the entire area covered by the chart. They are used for situations that are too complex or too unusual to be portrayed by standard symbology.

b. Cautions referring to a specific geographic position require a legend in the general location of the danger to direct the mariner to the text of the caution in the margin. If a single caution appears on the chart the legend shall read "SEE CAUTION." If multiple cautions are shown on the chart, they are numbered and the legends read "CAUTION NO 1, CAUTION NO 2, CAUTION NO 3," and so on. Note that the "SEE" is not shown for multiple cautions, and that there is no period after the abbreviation "NO"

c. The text of each caution is shown in the margin of the chart. Single cautions are not numbered and the caution title reads "CAUTION." If multiple cautions are shown, each caution is numbered and the caution title reads "CAUTIONS." The caution text in the margin is surrounded by a 0.3 mm lineweight box.

d. The caution box is positioned in the chart margin where space is available. Type for the caution legends is 12 point Swiss 742 upper case. Type for the caution title is 9 point Swiss 742 upper case. Type for the text of the caution notes is 7 point Swiss 742 upper and lower case. All caution information is shown in green (SPC-52813).

e. General notes are used to present information specific to the chart, not concerned with dangers to navigation. They are treated the same as cautions, with the exception that they are shown in black instead of green, and the legends "SEE CAUTION, CAUTION NO 1, CAUTIONS, etc." are replaced by the legends "SEE NOTE, NOTE NO 1, NOTES, etc."

3.11.7 Chart numbers. Chart Numbers are shown on each chart; outside the upper left, and lower left in black (SPC-58600). A chart number is not placed in the lower right since the Bar Code is placed there. Chart numbers are in 30 point Swiss 742.

3.11.8 Classification explanation. Classification Explanations are shown on all classified charts in black (SPC-58600) in 8 point Swiss 742, upper case. They are located in the lower right chart margin, 3mm below and centered under the Declassification Note. For charts whose classification is based exclusively on the location and existence of the chart, the standard note is shown (see style sheets, Appendices A and IB). If classified information is shown on the chart, the note is tailored to indicate the classified content. An example would be:

WARNING NOTICE - SECURITY CLASSIFICATION IS BASED ON TOPOGRAPHY (SECRET), HYDROGRAPHY (CONFIDENTIAL), AND THE FACT OF EXIXTENCE AND AVAILABILITY OF THIS SHEET. TO ENABLE UNCLASSIFIED REQUISITIONING, THIS SHEET MUST BE ORDERED BY STOCK NUMBER.

3.11.9 Classification notes.

a. Classification notes are shown on each classified Combat Chart in black (SPC-58600). Classification will be either SECRET or CONFIDENTIAL.

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b. Classified combat charts carry the classification in the following four places in the chart margin in 30 point Swiss 742, upper case.

(1) Between the seal and the chart title block (North-South) or above the chart title block (East-West).

(2) In the upper left margin, next to the chart number.

(3) In the lower right chart margin, next to the subtitle.

(4) In the left lower chart margin, next to the user's note.

c. Unclassified Combat Training Charts have a limited distribution and shall show the following note in place of the classification note, in positions b, c, and d described above. No note is shown in position 3.11.9.a.

LIMITED DISTRIBUTION

FIGURE 1. Classification Notice.

d. Type for the Limited Distribution is 18 point Swiss 742, upper case.

3.11.10 Source data.

a. Source Data is shown in black (SPC-58600) on all charts. The Source Data is shown to the right of the Accuracy Diagram on East-West oriented charts and below the Accuracy Diagram on North-South oriented charts. The Source Data lists topographic and hydrographic sources used to compile the chart, along with the date and scale of each source. Survey data is also provided for each hydrographic source. Source Data is shown in 6 point Swiss 742 upper and lower case type.

b. If known, survey dates of U.S. and foreign nautical chart sources shall be identified in the source listing associated with the accuracy diagram as shown in the examples below.

```
*     A-  Andalusian Chart 9197.................1:40,000
          (Ed. 1791, corr. 1968, from Spanish Navy surveys
          from 1776 to 1965)
      B-  N.O. Chart 01492.....................1:74,500
          (Ed. 1964, from U.S. Navy surveys in 1962)
      C-  Phenician Chart 0604.................1:141,000
          (Ed. 1959, from Phenician surveys to 1955)
      With additions from other sources
      * Uncorrected soundings
```

FIGURE 2. Example of a multi-source Source Data Note.

c. When listing sources, the only country name that shall be abbreviated is U.S. All other country names shall be spelled out. "British Admiralty" shall be spelled out when citing chart source material, and "British" shall be used when citing surveys, originating from the Hydrographic Department, Ministry of Defence of the United Kingdom.

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d. If original surveys are cited instead of charts, they shall be listed by their archive number, if known. DMA document numbers shall not be shown as a substitute for the producer's archive number. Surveys done by private companies shall be cited as "Commercial Surveys" without a number. Separate sheets of the same survey are not listed separately, or delineated separately in the source diagram. However, if separate sheets are at different scales, the largest and smallest scales shall be listed as a range. Dates of separate sheets shall be shown in the same manner.

e. If sources cannot be corrected for sound velocity they shall be identified in the compilation data list. A note shall be added at the bottom of the list of hydrographic sources. It shall read: " * Uncorrected soundings." An asterisk shall be added to both the source reference line, as shown in the example above, and to the corresponding source identifier in the accuracy diagram.

f. If all soundings on a chart are either corrected for sound velocity or uncorrected, the corrected sounding note is not required in the compilation data list and the presence or absence of the corrected sounding note in the vertical datum note shall be used to indicate that soundings are corrected for sound velocity, as described in 3.11.31.a.(4).

3.11.11 Datum note.

a. Datum notes are shown on Combat Charts that are not constructed on the World Geodetic System 1984 (WGS-84) datum. The datum note indicates the necessary correction required to place the chart on WGS-84. The datum note is based on one of the following conditions.

(1) If a shift can be derived from the chart datum to WGS, the following note shall be shown:

COORDINATE CONVERSION (datum name) TO WGS

GRID: Subtract 65 m E; Subtract 296 m N
Geographic: Subtract 3.5" Long; Subtract 3.0" Lat.

Note that the grid and geographic coordinates and the directions shown above are examples and will vary from chart to chart.

(2) If the chart cannot be shifted to WGS because of a lack of geodetic data or if the chart is internally inconsistent, the following note shall be shown:

WORLD GEODETIC SYSTEM DATUM ADJUSTMENT

Due to the unavailability of geodetic data, this chart cannot be placed on the World Geodetic System 1984 (WGS-84) Datum.

(3) When the chart is constructed on the World Geodetic System 1984 (WGS-84), no datum note is required.

b. These notes are shown in black (SPC-58600) in the upper right chart margin on North-South oriented charts, with the right end aligned with the right chart border, and in the top margin on East-West oriented charts. Type is 9 point Swiss 742 upper case for the title and 7 point Swiss 742 upper and lower case for the text.

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3.11.12 Declassification notes. Declassification Notes are shown on all classified charts in black (SPC-58600) 8 point Swiss 742, upper case type. They provide the authority for classification and downgrading instructions for the charts. They are located in the lower margin, 3mm below and centered on the Classification Note in the right lower margin.

3.11.13 Depth conversion scale. A standard Depth Conversion Scale showing the relationship between feet, meters and decimeters, and fathoms and feet is shown on each chart in black (SPC-58600). It is located in the lower chart margin on all charts.

3.11.14 Disclaimer note. A disclaimer note, if applicable, is shown in the chart margin to indicate that names and boundaries are not necessarily authoritative. This note appears in black (SPC-58600). The type is 8 point Swiss 742, upper and lower case. The disclaimer note is not shown on charts of U.S. domestic areas.

3.11.15 DMA Seal. The DMA Seal is shown on each chart. The standard 19mm Defense Mapping Agency Seal is shown in black (SPC-58600) above the title block on North-South charts and in the top margin on East-West charts.

3.11.16 DMA bar code and stock number. The DMA Stock Number is shown within the alpha-numeric code of the Bar Code on all charts and placed on the lower right corner of the chart, centered 5 mm below the chart number. The DMA Bar Code, with the DMA Stock Number, is shown in accordance with MIL-STD-600010. The first five letters of the stock number are COMBT, followed by the six digit chart number.

3.11.17 Edition number and date. The edition number and date are shown in black (SPC-58600). The edition date is the Saturday date of the latest Notice to Mariners checked for changes on source charts. All first and subsequent chart editions will show edition number and date followed by the note:

Correct through NM _/ _

This note is placed below the lower left corner on all charts, in 8 point Swiss 742, upper and lower case type. Classified Combat Charts will be referenced to the Classified Notice to Mariners, and carry the note:

Correct through CN _/ _

3.11.18 First edition date. The First Edition Date identifies the earliest date that the chart was published and gives an approximation of the date of original compilation. The first edition date is shown on each chart in black (SPC-58600) in 7 point Swiss 742 type. The First Edition Date is shown below and centered on the Horizontal Datum Note.

3.11.19 Glossary. The glossary is shown in black (SPC-58600) and located in the right chart margin on North-South oriented charts, and in the top chart margin on East-West oriented charts. When the native language is other than English, all generic terms appearing on the chart are translated in the glossary. Since one glossary may be utilized for a group or series of charts, all items appearing in the

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glossary may not apply to any one chart. The heading is shown in 9 point Swiss 742, upper and lower case type and the translations are in 7 point Swiss 742, upper and lower case type.

3.11.20 Grid reference box.

a. The grid reference box is shown on each chart. It contains instructions and an example for composing a standard grid reference. The grid reference box also contains diagrams identifying the applicable grid zone designations and grid square identifications. See DMA Technical Manual 8358.1, "Datums, Ellipsoids, Grids, and Grid Reference Systems."

b. The grid declination from true north for each grid appearing on the chart is shown for the approximate mid-latitude of the east and west chart edges.

c. The grid reference box and instructions are shown in purple (SPC-96532). The sample point is shown as it appears inside the neatline. When only the primary grid is shown, the grid zone designation and 100,000 meter square identification letters are also shown in purple. When there is a grid zone junction on or near the chart, and primary and secondary grids must be shown, the grid zone designation is shown in black (SPC-58600), and the 100,000 meter square identification letters are shown in the grid color. Appendix I (East-West Style Sheet) shows an example of a grid reference box when two grids are shown. Appendix II (North-South Style Sheet) shows an example of a grid reference box when only the primary grid is shown.

3.11.21 Heights note.

a. A Heights Note, indicating the unit of measure for heights, is shown on each chart. The first line of the note "HEIGHTS IN METERS"; is shown in black (SPC-58600) in 14 point Swiss 742, upper case type. Notes indicating the contour interval and reference to supplementary contours, if required, are shown in red-brown (SPC-61121) under the first line. Contour information is in 8 point Swiss 742, upper and lower case, and supplementary contour information is in 7 point Swiss 742, upper and lower case type.

b. The information contained in the Heights Note is repeated in the Vertical Datum Note and these notes must be in agreement regarding contour interval, and supplementary contour interval if applicable.

3.11.22 Horizontal datum note.

a. Horizontal Datum notes are shown on each chart and contain the following information:

(1) Type of projection - Transverse Mercator

(2) Horizontal datum - see below

(3) Grid interval - 1000 meters

(4) Grid zone and ellipsoid - see below

(5) Scale - 1:50,000

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b. The horizontal datums, grid zones, and ellipsoids used on Combat Charts are those specified in the latest edition of DMA Technical Manual 8358.1, "Datums, Ellipsoids, Grids and Grid Reference Systems."

c. Type is Swiss 742, upper case. The type of projection is in 10 point, the rest of the Horizontal Datum note is 8 point. The type of projection, datum, and scale are shown in black (SPC-58600). UTM primary grid information is shown in purple (SPC-96532), and any UTM secondary grid information, if required, is shown in blue (SPC-47651). Occasionally a third grid is required; either a local grid or a grid crossing datum/ellipsoid boundaries. If the third grid is required, it and all related notes are shown in Red-Brown (SPC-61121).

3.11.23 Location diagram.

a. A Location Diagram is shown on each chart. The diagram portrays the location of the chart in relation to other charts in the series by a heavy outline of the chart. Degree graticule lines are shown and labeled. Shoreline and major names (country, city or town and water area identification) are shown. The diagram is black (SPC-58600), except for water area names, which are blue (SPC-47651), and water areas, which are tinted blue (SPC-47651, 21% screen). Location Diagrams are not shown for isolated Combat Charts.

b. The diagram is located below the title block in the right chart margin on North-South oriented charts and in the top chart margin on East-West oriented charts. Land names are shown in Swiss 742 type; country names are in 10 point, upper case, city or town names are 8 point, upper and lower case; chart numbers are 8 point Swiss 742 type, and water area names are 10 point Swiss 742 italic type.

3.11.24 Main title.

a. The main title is shown on each chart. It includes the following information:

(1) Type of chart - either COMBAT CHART or COMBAT TRAINING CHART

(2) Major water name - geographic name of ocean or sea

(3) Country name - name of nation

(4) Chart title - geographic name of specific locality

3.11.25 Publishing note. The DMAHTC Publishing Note identifies DMAHTC as the publishing authority. It is shown on each chart in black (SPC-58600) and is centered between the right and left neatlines in the lower chart margin. The first and third lines are 8 point Swiss 742 upper and lower case type; The second line is upper case only.

3.11.26 Sounding note. The unit of measurement used for soundings appears in purple (SPC-96532) and is located in the top and bottom margins on all charts. The type is 30 point Swiss 742, upper case.

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3.11.27 Subtitle.

a. The Subtitle is shown on each chart. It identifies the following information:

(1) Type of chart - either COMBAT CHART or COMBAT TRAINING CHART

(2) Chart title - geographic name of specific locality. This name must be the same as the one in the main title.

(3) Unit of measure for soundings - Meters

(4) Scale - 1:50,000

b. The subtitle is shown in black (SPC-58600). The type of chart is in 10 point Swiss 742 Light Condensed, upper case type, the chart title is in 14 point Swiss 742 Light Condensed, upper and lower case type, and unit of soundings and scale is in 8 point Swiss 742 Light Condensed, upper case type. It is located below the lower right chart corner.

3.11.28 Symbol legend.

a. A Symbol Legend is shown on each chart. The Symbol Legend defines and illustrates the features most frequently represented in a series of charts. It is the same legend that is used on the equivalent scale topographic maps of the same area, with additional hydrographic features added. These features are light 2C050 (Replaces the topographic symbol for lighthouse) , sunken rocks 2D130 VRC=4, rocks awash 2D130 VRC=2, foreshore 2A020, sunken wreck 2D180 VRC=4, exposed wreck 2D180 VRC=1, Danger 2D000 SFC=2 without a label, reef 2D120 VRC=2, depth curve 2E010 CRV=10, and sounding 2E020 SND=6 HDP=3.

b. The legend shall be bi-lingual when the equivalent scale topographic maps are shown with a second language. The English descriptive type shall be vertical and the foreign language type shall be italic.

c. The legend is located below the Glossary on North-South oriented charts and in the top margin on East-West oriented charts. The legend title is shown in 10 point Swiss 742, upper case type. The names of the symbols are in 5 point Swiss 742, upper case type (for major items), or upper and lower case type (for sub-items). Color for this type is black (SPC-58600).

d. The symbols and symbol labels in the legend are shown the same as they appear inside the chart neatlines and are specified in Section 3 of MIL-STD-600002.

3.11.29 Tide box.

a. A Tide Box is shown when sufficient data is available. Tidal information includes the name of the tidal stations located on the chart or the most representative tidal station in the proximity, and the height of tides in relation to the datum of soundings. The height of tides is indicated by gradients of Higher and Lower Spring and Neap

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tides or mean Ranges as appropriate. Place names and tidal heights shown below are examples.

(1) Diurnal tides are tides with a single high and a single low water each day. Tide boxes shall show the tide levels of the mean heights of high and low water. The positions in the tide box for the Mean Lower High Water and the Mean Higher Low Water are left blank. A statement indicating the type of tide shall be included.

TIDAL INFORMATION

Place	Height above datum of soundings				
	Mean High Water		Mean Sea Level	Mean Low Water	
	Higher	Lower		Lower	Higher
	meters	*meters*	*meters*	*meters*	meters
Abadan	1.3	--	0.7	0.1	--
Karg Island	1.1	--	0.5	0.0	--

Tide is usually diurnal

FIGURE 3. Tidal Box for diurnal tides.

(2) Semi-diurnal tides are tides with two high waters and two low waters each day, and the difference between the two high waters and the differences between the two low waters is small. The tide box shall show the mean heights of the high and low water at both springs and neaps.

TIDAL INFORMATION

Place	Lat	Long	Height above datum of soundings				
			Mean High Water		Mean Sea Level	Mean Low Water	
			Higher	Lower		Lower	Higher
			meters	meters	meters	meters	meters
Jutland	55° 42'N	008 °14'E	1.5	1.3	0.8	0.1	0.4
Helgoland	52° 22'N	007°46'E	1.4	1.1	0.7	0.0	0.2

FIGURE 4. Tidal Box for semi-diurnal tides.

(3) Mixed tides are tides with two high waters and two low waters each day with a large difference between the heights of the highs and/or a large difference between the heights of the lows. Tide boxes shall show the mean heights of the two daily high and low waters.

TIDAL INFORMATION

Place	Height above datum of soundings				
	Mean High Water		Mean Sea Level	Mean Low Water	
	Higher	Lower		Lower	Higher
	meters	meters	meters	meters	meters
Tokyo Wan	1.3	0.8	0.5	0.1	0.4
Yokosuka	1.1	0.5	0.3	0.0	0.1

FIGURE 5. Tidal Box of mixed tides.

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b. If full information is not available, partly complete data, for example, a height for springs only, may be given. Other spaces are left blank.

c. Figure 4 shows an example of a tide box with geographic positions. The geographic coordinates of tide stations (to the nearest minute) are given when one or more of the following conditions exist:

(1) The tide station listed in the box falls outside the limits of the chart. Tide stations that are less than nine kilometers (five miles) outside the limits of the chart may be used if no appreciable difference in tide height will result.

(2) The tide station cannot be readily located on the chart.

d. Tidal information is shown in black (SPC-58600) and located in the lower left chart margin on East-West oriented chart and in the right chart margin on North-South oriented charts. All type is 7 point Swiss 742 caps and lowercase, with the exception of the title, which is 7 point Swiss 742 caps.

e. When no tidal information is available for a chart, the following note is shown in the space specified above for tidal information:

TIDAL INFORMATION

There is no tidal information available for the area covered by this chart.

3.11.30 User's note. The DMAHTC User's Note provides instructions for users to provide comments and suggestions for improving the chart. The note appears in black (SPC-58600) and is located in the bottom left margin below the edition number, edition date and chart number on all charts. The type is 6 point Swiss 742, upper and lower case. See following example:

USERS SHOULD REFER CORRECTIONS, ADDITIONS,
AND COMMENTS FOR IMPROVING THIS PRODUCT
TO: DIRECTOR, DEFENSE MAPPING AGENCY, ATTN:
PR, 8613 LEE HIGHWAY, FAIRFAX, VA 22031-2137.

3.11.31 Vertical datum notes.

a. Vertical Datum notes are shown on each chart and contain the following information:

(1) Unit of soundings - meters.

(2) Unit of soundings to greater precision - identifies the depth to which soundings are shown in meters and decimeters.

(3) Sounding datum - identifies the vertical datum to which depth information is referenced. When the vertical datum for depths is a low water datum the note is "reduced to the approximate level of (datum name)." When the vertical datum for depths is Mean Sea Level, the note is "reduced to approximate Mean Sea Level."

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(4) Sounding correction note - indicates that the soundings on the chart have been corrected for sound velocity. The absence of this note indicates that not all or none of the the soundings have been corrected for sounding velocity

(5) Unit of measure for heights - meters

(6) Contour interval - identifies the contour interval

(7) Supplementary contour note - indicates that the supplementary contours are shown as dashed lines and provides the supplementary contour interval. This note is shown only when supplementary contours are shown on the chart.

b. Type is Swiss 742. Unit of soundings is in 10 point, upper case, soundings differing from the main unit soundings are in 8 point, upper and lower case, sounding datum is in 8 point, upper and lower case, height units and height datum are in 10 point, upper case, contour intervals are in 8 point, upper and lower case, and supplementary contours are in 7 point, upper and lower case. The Vertical Datum note is shown in black (SPC-58600), except the contour and supplementary contour intervals which are shown in red-brown (SPC-61121).

3.11.32 Warning note. One of the following warning notes, as appropriate, is shown in the chart margin in purple (SPC-96532). The type for the title is 9 point Swiss 742, upper case, and the type for the text is 7 point Swiss 742, upper and lower case:

a. For charts of foreign waters the warning reads:

WARNING

The prudent mariner will not rely solely on any single aid to navigation, particularly on floating aids. See paragraph No. 1 of Notice to Mariners No. 1 or Sailing Directions Planning Guides for information relative to DMAHTC Charts.

FIGURE 6. Warning Note for charts of foreign waters.

b. For charts of U.S. waters the warning reads:

WARNING

The prudent mariner will not rely solely on any single aid to navigation, particularly on floating aids. See U.S. Coast Guard Light List and U.S. Coast Pilot for details.

FIGURE 7. Warning Note for charts of U.S. waters.

3.12 Culture.

a. Combat Charts provide a combination of hydrographic chart and topographic map for use in amphibious operations. Therefore, the cultural information shown on Combat Charts is the same as shown on the

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1:50,000 scale topographic maps, as specified in MIL-T-89301 (latest edition), with the exception of certain marine aids to navigation.

b. Combat Training Charts shall show the same information that is shown on 1:50,000 Combat Charts.

c. Certain marine aids to navigation are shown on Combat Charts differently than on topographic line maps. Refer to Category 2C code features in 3.24 of this specification for specific feature requirements.

(1) Operational lighthouses are shown as marine navigational lights (2C050) rather than as buildings (1L015).

(2) Other marine lights and beacons are shown on Combat Charts even though they are not shown at all on topographic maps.

3.13 *Hydrography*.

a. The hydrography is compiled from the latest and most reliable hydrographic survey data available. Due to the sparsity of hydrographic surveys and the normal inaccessibility of foreign survey smooth sheets, the most frequently utilized source materials are published nautical charts.

b. The unit of measure for all hydrographic data is meters. Depths under 21 meters are shown in meters and decimeters if source data is available.

c. Soundings shown on Combat Charts shall be corrected for sound velocity where possible. Soundings shallower than 200 meters shall be corrected whenever bar checks, sound velocity, or temperature/salinity data are available (SVC=4). Soundings 200 meters or deeper shall be corrected using Echo-Sounding Correction Tables, NP 139 (latest edition) issued by the British Admiralty Hydrographic Department (SVC=3). If directly observed sound velocity data are available, they should be used if considered to be more reliable than the average correction values for the zones in NP 139.

d. Charts showing only corrected soundings (SVC=3 or 4) shall show a note with the other vertical datum notes as specified in 3.11.31.a.(4). This note shall be omitted from charts that show soundings that have not been corrected for sound velocity.

e. If it is not possible to correct the soundings from sources used on a Combat Chart, and therefore a mixture of corrected and uncorrected soundings are shown, those sources with uncorrected soundings shall be identified in the compilation data list as described in 3.11.10.

f. Consult IHO Publication No. MP004 (latest edition) or the producing agency to determine if soundings on foreign source charts require sound velocity corrections. Soundings on most foreign charts have already been corrected for sound velocity.

g. Sources in fathoms and feet that have been corrected for sound velocity shall be converted to meters using Standard Conversion Table No. 4. Sources in fathoms and feet that have not been corrected

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for sound velocity and have an assumed speed of sound in salt water of 1500 meters per second (820 fathoms per second) shall be converted to meters using Conversion Table No. 4 before sound velocity corrections are applied. Sources in fathoms and feet that have not been corrected for sound velocity and have an assumed speed of sound in salt water of 1463 meters per second (800 fathoms per second) shall be converted to meters using Standard Conversion Table No. 3 before sound velocity corrections are applied. See Publication NP 139.

h. Hydrography on Combat Charts is portrayed primarily by soundings, and supplemented by depth curves. The soundings and depth curves must present a representation of the bottom that will allow a mariner to interpolate the depth of water at places where soundings and depth curves are not shown, with some degree of accuracy. Therefore, the selection of soundings to be shown on the chart is critical to the usability of the chart. Rules for sounding selection are presented in 3.24 of this specification, but rules cannot cover every conceivable situation. The judgement of an experienced nautical cartographer must be the final authority on the correct selection of soundings.

i. Soundings can be classified as either critical soundings, support soundings and fill soundings. Critical soundings are those soundings which identify the shoals and soundings which provide essential information required for navigation in non-dangerous areas. Support soundings provide additional information about the shape of the bottom around critical soundings and shoals, or are used to provide identifiers to depth curves and to show changes in bottom slope away from shoals or deeps. Fill soundings are used to fill in flat areas and to fill in deep areas that are not adequately defined by support soundings.

j. Emphasis must be placed on selecting a density of soundings for natural channels, shoals or other hazardous areas that are sufficiently close together so that these areas are properly highlighted (by dense sounding pattern) for quick recognition by the mariner. Additional supportive soundings and fill soundings are selected at a lesser density to complete the bottom description.

k. To support ship to shore movement, the hydrographic detail inside the 10 meter depth curve must be as detailed as possible. Sounding density in this area is greatest, and all dangers and bottom characteristics must be shown. Depth curves at 1, 2, and 3 meters should be shown if sounding data is sufficient to support them.

l. The depth curve interval may be modified to better fit the configuration of the bottom. Depth curve interval may be increased in areas with a steeply sloping bottom, and decreased in areas of gently sloping bottom. Depth curve interval is also dependent on the source material. Depth curves can be interpolated at the desired interval from hydrographic surveys but when published nautical charts are used as the primary source depth curve interval will be dependent on what is shown on source charts.

3.14 Hypsography/Physiography.

a. The hypsography and physiography are generally portrayed on Combat Charts the same as on equivalent scale topographic maps, MIL-T-89301.

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b. Tints are used to emphasize land and water areas of the chart. A yellow tint (SPC-57437, 31% 15° angle screen) is printed over land areas of the chart, and land areas of all applicable diagrams in the chart margin. Land tint is also printed in offshore features (outside the shoreline) that are above the high water plane of reference (VRC=1), such as breakwaters, piers, cribs, or offshore loading facilities. Land tint is not printed over glaciers or snowfields, which have a white background.

c. Water tint (SPC 47651, 21%, 45° angle screen) is used to emphasize water areas that are less than 10 meters deep. Water tint is shown from the shoreline to the 10 meter depth curve, and overprints the foreshore (2A020), features that cover and uncover with the rise and fall of the tide (VRC=8), and features that are entirely under water (VRC=4). The blue-gray tint printed in area features that cover and uncover with the tide is created by the open water tint overprinting the 12% black tint specified in the feature symbols.

3.15 Vegetation. Vegetation is portrayed on Combat Charts the same as on equivalent scale topographic maps, MIL-T-89301.

3.16 Demarcation.

a. Boundaries and identification of administrative divisions shall be in accordance with current U.S. Department of State policy.

b. Portrayal of international and first order administrative land boundaries on Combat Charts are portrayed the same as on the equivalent scale topographic maps, MIL-T-89301. The Boundaries Graphic in the margin of topographic maps is not shown on Combat Charts. Political boundaries shall not be shown in the water.

c. Hydrographic demarcations are generally limited to those features that affect safety to surface navigation. Boundaries for other regulated areas, such as fishing prohibited areas or traffic separation schemes, are not shown because they would not be in force or of interest to forces conducting amphibious operations in wartime.

d. Hydrographic demarcations on Combat Charts are symbolized in black or green. Purple symbols are not used because they would visually conflict with the purple UTM grid lines.

3.17 Aeronautical. This section is not applicable to this specification.

3.18 Names and labeling.

a. Refer to MIL-STD-600002, MIL-STD-600003 and MIL-STD-600004 for proper naming and labeling of applicable features.

b. The following is a list of features which may not appear in 3.24.2, but may be named on the final product. Definitions for the following features may be found in MIL-STD-600004.

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Name	Example
Banks	Outer Banks
Basin	Great Basin
Bay	Chesapeake Bay
Beach	Virginia Beach
Bench	
Bend	
Bluff	
Bottom	
Break	
Butte	
Canyon	Grand Canyon
Cape	Cape of Good Hope
Channel	English Channel
City	New York City
Cliff	
Corner	Tyson's Corner
Cove	
Crossing	
Desert	Sahara Desert
Dispersed Village	
Dome	
Everglade	Florida Everglades
Falls	
Flat	
Flats	
Forest	
Gap	
Gorge	
Gulch	
Gulf	Gulf of Mexico
Gut	
Hamlet	
Harbor	Boston Harbor
Head	
Highland	
Hill	
Hole	
Hollow	
Inlet	Hamilton Inlet
Hollow	
Inlet	Hamilton Inlet
Island Chain	Hawaiian Islands
Junction	
Jungle	
Knob	
Knoll	
Lagoon	
Lake	
Lands	
Lookout	
Mesa	
Mountain	
Marina	
Mountain Range	Rocky Mountains
Narrows	
Neck	

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Name	Example
Ocean	Atlantic Ocean
Park	Yellowstone National Park
Pass	
Passage	
Patch	
Peak	Pikes Peak
Plain	Great Plains
Plateau	Colorado Plateau
Point	
Pool	
Port	
Range	Coastal Range
Ravine	
Region	
Ridge	
River	
Roadstead	
Rock	
Sands	
Scattered Village	Comunidades of South America, Strevsudlung of Europe
Sea	Caribbean Sea
Sea Mount	
Shelf	
Shoals	
Sink	
Sound	Puget Sound
Spit	
Spring	
Spur	
Strait	Bering Strait
Summit	
Town	
Valley	Death Valley
Village	Greenwich Village
Wood	

c. The names on Combat Charts shall be the same names shown on the equivalent scale topographic maps, PS/3AA/190 (latest edition). If possible, names and labels for land features shall not be placed in the water where they might obscure hydrographic detail. Geographic names on offshore features should be the same as shown on hydrographic charts covering the same area. Named features that are described in the Sailing Directions publications should also be named.

3.19 Radar. No radar information is required on Combat Charts.

3.20 Intelligence information. This section is not applicable to this specification.

3.21 Special areas. This section is not applicable to this specification.

3.22 Symbology. Symbology for Combat Charts shall be in accordance with MIL-STD-600002.

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3.23 Reproduction.

a. Reproduction of Combat Charts is by lithography. The final product conforms to the best lithographic quality standards with respect to clarity, conformance to specified colors and screens, and accuracy of registration.

b. Combat charts are printed on high wet strength, E20, Task Force Chart paper. The standard trim size is 105.4 cm (41 1/2 inches) by 147.3 cm (58 inches) for all Combat Charts (North-South and East-West).

c. Printing colors and screens shall conform to information and items illustrated in MIL-STD-600005.

3.24 Feature/Attribute.

3.24.1 General. This section contains feature, feature attributes category, feature attribute category value, inclusion condition and specific rules corresponding to Combat Chart production.

3.24.2 Feature/Attribute category, inclusion conditions and product generation rules. The following is an explanation of the header format for Table I:

FCode (1)	Feature (2)	FT (3)	ACode (4)	Attribute (5)	Inclusion Condition (6)	Rule (7)

(1) F (Feature)Code - 5 digit alpha numeric, Feature Attribute Coding Standard (FACS) Code assigned to each feature (e.g. 1N010 - R/R Tracks). The first two digits identify the category and subcategory to which each feature belongs (e.g., 1 - Culture Category, N = Transportation R/R subcategory).

(2) Feature - Name of feature as specified in the FACS. A feature is a physical (e.g., Bridge) or conceptual (e.g., Route - Nautical) entity of the real world which has one or more set of coordinates to be included on a product.

(3) Feature Type - designation of a feature type.

Area - More than two sets of coordinates defining a closed area; areas may span more than one map sheet or geographic area requirement.

Line - Two or more coordinate sets defining a series of line segments.

Point - One set of coordinates.

If there is more than one Feature Type for the feature, then the ACode and Inclusion conditions are stated separately for each type.

(4) A(Attribute)Code - Three digit alpha or alpha numeric character (acronym) FACS code assigned to each attribute category which identifies the attribute category (e.g., EXS - Existence Category). Attribute categories are defined by mutually exclusive sets of attribute

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values which are feature dependent. Attribute values relative to product are normally contained in MIL-STD-600002 under column headed "SValue", a few exceptions are contained in the inclusion conditions.

(5) Attribute - Name of attribute category required by the feature as specified in the FACS. Attribute categories are characteristics in menu form relative to a specified feature or features.

(6) Inclusion conditions - Conditions under which the feature/attribute(s) are required by the product (e.g., R/R Yard, 1N080 FAC Code, is included on a particular product only if Length >= 450m). Conditions should be stated in boolean logic.

(7) Rule - 5 digit alpha-numeric code indicating rules (listed in MIL-STD-600003) which specify requirements for feature to satisfy final product format/requirements.

3.25 Magnetic variation.

3.25.1 Compass rose portrayal. Magnetic variation data is shown on Combat Charts by means of black (SPC-58600) compass roses oriented to true north and a legend in the center of each compass rose indicating the magnetic variation (to the nearest minute) at the position of the compass rose, and annual change (to the nearest second). Refer to the Style Sheets, Appendices A and B, for graphic examples of compass roses.

a. In rounding off to nearest minutes and seconds, 29 minutes and 29 seconds are rounded down, and 30 minutes and 30 seconds are rounded up.

b. Magnetic variation data is shown for the five year epoch in which the edition date falls. For example, a chart published in 1993 shows magnetic data for the epoch year 1990. Magnetic variation data shall not be updated for intermediate years.

3.25.2 Compass positions. The positions of compass roses are chosen to meet the following criteria:

a. Placed in the water areas of the chart in such a way that they do not obscure dangers, shoal soundings, or aids to navigation.

b. Placed in the vicinity of the meridians which are central to the areas where the compass roses will most likely be used. This minimizes the error introduced by the convergence of meridians on the Transverse Mercator projection.

c. Clear of chart folds.

d. Provide coverage over the chart so that no water area, excluding inland hydrography, is farther than 45 cm from a compass rose.

3.25.3 Compass rose specifications.

a. A compass rose consists of 360 ticks radiating out from the center of the compass rose. The ticks begin at a radial distance of 62 mm from the center of the compass rose. Tick lengths are as follows:

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10° ticks are 3.0 mm long, 5° ticks are 2.0 mm long, and 1° ticks are 1.0 mm long. All lineweights are 0.15 mm.

b. Ten degree ticks (0, 10, 20, 30, 40,...350) are labeled with 7 point type, with the numbers oriented so they are reading from the bottom of the chart (90 and 270 are vertical and reading from the inside of the compass rose). No degree signs are shown. Space between the end of the 10° ticks and the base of the type is 1.0 mm.

c. In addition to these ticks, extended ticks are shown at 0°, 90°, 180° and 270°. These ticks are 5.0 mm long and begin 1.0 mm outside of the numbers 0, 90, 180 and 270.

d. A five pointed star is shown above the tick over the number 0. The star is 7.5 mm in diameter and is shown by a perimeter line (0.15 mm lineweight). Each point of the star is then divided by another line down the centerline of the point (0.15 mm lineweight). The right half of each point (proceeding clockwise around the star) is shaded by lines, parallel to the center line and spaced at 0.2 mm interval (lines are 0.10 mm lineweight). The star is positioned so one of the points is pointing north and the end of the tick above the number "0" is touching the perimeter of the star opposite the top point.

e. The cross at the center of the compass rose is 0.15 mm lineweight. Ticks are oriented 90 degrees to each other and are positioned at cardinal directions (north-south, east-west). The overall tick length is 2.0 mm.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Final product quality. Final product quality will reflect the quality elements expressed by each appropriate section within MIL-STD-600000, MIL-STD-600001, MIL-STD-600002, MIL-STD-600003,MIL-STD-600004 and MIL-STD-600005.

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5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. Combat Charts are 1:50,000 scale charts used for plotting amphibious landings and firing direction from ships. Combat Charts are produced to support the U.S. Marines.

6.2 Aquisition requirement. When this specification is used in aquisition, the applicable issue of the DODISS must be cited in the solicitation (see 2.1.1).

6.3 Supersession. These specifications supersede Defense Mapping Agency Product Specifications for Combat Charts, PS/2AD/040, December 1984.

6.4 Definitions.

6.4.1 Circular error (CE) - An accuracy figure representing the stated percentage of probability that any point expressed as a function of two linear components (e.g., horizontal position) will be within the given figure.

6.4.2 Linear error (LE) - A one dimensional error (such as an error in elevation) defined by the normal distribution function.

6.5 International standardization agreements.

"Certain provisions of this specification are subject of international standardization agreement. When amendment, revision, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations."

6.5.1 International Standardization Agreements (STANAGs)

This section is not applicable to this specification.

6.5.2 Quadripartite Standardization Agreements (QSTAGs).

This section is not applicable to this specification.

6.5.3 Air Standardization Coordinating Committee Agreements (ASCC AIR STDs/STDs/ADV PUBs.

This section is not applicable to this specification.

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6.5.4 International MC&G Agreements.

This section is not applicable to this specification.

6.5.5 Executive Orders.

This section is not applicable to this specification.

6.5.6 Inter-Agency Agreements.

This section is not applicable to this specification.

6.5.7 Other Documentation.

This section is not applicable to this specification.

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: EXTRACTION (1A)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1A010	MINE			
Area	ARA	Area Coverage Attribute	(EXS 0 or 28)	L-3801
	EXS	Existence Category	and >= 15,625 m square ARA	L-4006
	LMC	Landmark Category	OR MIN 8	L-4007
	MIN	Mining Category	and >= 15,625 m square ARA	L-4008
	NAM	Name Category	OR EXS 6	L-4010
	PRO	Product Category	and LMC 1	R-2244
			and >= 15,625 m square ARA	R-2369
				R-2494
Point	ARA	Area Coverage Attribute	(EXS 0 or 28)	D-1653
	EXS	Existence Category	and < 15,625 m square ARA	L-3801
	LMC	Landmark Category	OR MIN 8	L-4006
	MIN	Mining Category	and < 15,625 m square ARA	L-4007
	NAM	Name Category	OR LMC 1	L-4008
	PRO	Product Category	and EXS 6	L-4010
			and < 15,625 m square ARA	O-3156
				R-2248
				R-2369
1A030	QUARRY			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	EXS	Existence Category	and (EXS 0 or 28)	L-4006
	LMC	Landmark Category	OR LMC 1	R-2243
	PRO	Product Category	and EXS 6	R-2494
			and >= 15,625 m square ARA	
Point	ARA	Area Coverage Attribute	< 15,625 m square ARA	R-2248
	EXS	Existence Category	and (EXS 0 or 28)	L-4006
	LMC	Landmark Category	OR LMC 1	O-3156
	PRO	Product Category	and EXS 6	D-1653
			and < 15,625 m square ARA	L-3801

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: EXTRACTION (1A)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1A040	RIG /SUPERSTRUCTURE			
Point	HGT	Height Above Surface Level	LOC 2	L-3801
	LOC	Location /Origin Category	OR LOC 9	L-3972
	PRO	Product Category	and >= 46 m HGT	L-4006
	ZVL	Z Value		R-3674
				R-0046
				O-3156
				T-0304
1A050	WELL			
Point	EXS	Existence Category	(PRO 0 or 12 or 18)	T-0300
	HYC	Hydrographic Category	and (EXS 0 or 3 or 6 or 28)	R-2244
	LMC	Landmark Category	OR PRO 27	R-2246
	NAM	Name Category	and (HYC 0 or 6 or 8)	R-2248
	PRO	Product Category	and (SCC 0 or 1 or 4 or 9)	D-1653
	SCC	Spring /Well Characteristic Category	OR PRO 27	L-4006
			and (EXS 0 or 3 or 28)	L-4008
	WFT	Well Feature Type	and HYC 8	L-4009
			OR PRO 27	L-3801
			and EXS 6	O-3155
			and HYC 3	O-3156
				L-4706
				L-4813

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: DISPOSAL (1B)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1B000	DISPOSAL SITE /WASTE PILE			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	LMC	Landmark Category		R-2243
	PRO	Product Category		R-2494
1B010	WRECKING YARD /SCRAP YARD			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	LMC	Landmark Category		R-2243
				R-2494
				R-3730
				R-3732
				R-3733

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: PROCESSING INDUSTRY (1C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1C000	PROCESSING PLANT /TREATMENT PLANT			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	NAM	Name Category		L-4006
	PRO	Product Category		L-4008
				L-4010
				L-4027
				R-2494
				L-4813
Point	ARA	Area Coverage Attribute	< 15,625 m square ARA	D-1653
	NAM	Name Category		L-3801
	PRO	Product Category		L-4006
				L-4008
				L-4010
				R-2248
				L-4813
1C020	CATALYTIC CRACKER			
Point	HGT	Height Above Surface Level	All required	D-1653
	ZVL	Z Value		L-3801
				R-2248
				R-0046
1C030	SETTLING BASIN /SLUDGE POND			
Area	WID	Width	>= 125 m WID	L-3801
				R-2494

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: POWER GENERATION (1D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1D010	POWER PLANT			
Area	ARA	Area Coverage Attribute	>= 40 m WID	L-3801
	HGT	Height Above Surface Level	and < 46 m HGT	L-4008
	LMC	Landmark Category		R-2243
	NAM	Name Category		L-4011
	PPC	Power Plant Category		L-4813
	WID	Width		
Point	HGT	Height Above Surface Level	< 40 m WID	D-1653
	NAM	Name Category	OR >= 40 m WID	L-3801
	PPC	Power Plant Category	and >= 46 m HGT	L-4011
	WID	Width		R-2248
	ZVL	Z Value		R-0046
				L-4008
				O-3156
				L-4813
1D020	SOLAR PANEL			
Point			All required	D-1653
				L-3801
				R-2248
1D030	SUBSTATION /TRANSFORMER YARD			
Area	WID	Width	> 40 m WID	L-3801
Point	LMC	Landmark Category	<= 40 m WID	D-1653
	WID	Width	and LMC 1	L-3801
				R-2248
				O-3156

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: ASSOCIATED INDUSTRIAL STRUCTURES (1F)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1F010 CHIMNEY /SMOKESTACK				
Point	HGT	Height Above Surface Level	>= 46 m HGT	R-0046
	LMC	Landmark Category	OR < 46 m HGT	O-3156
	ZVL	Z Value	and LMC 1	D-1653
				R-2248
				L-3801
1F020 CONVEYOR				
Line	LEN	Length /Diameter	>= 375 m LEN	L-3801
	LMC	Landmark Category	OR LMC 1	R-2331
				O-3156
1F030 COOLING TOWER				
Point	HGT	Height Above Surface Level	>= 46 m HGT	D-1653
	LMC	Landmark Category	OR < 46 m HGT	R-2248
	ZVL	Z Value	and LMC 1	L-3801
				O-3156
				R-0046
1F040 CRANE				
Point	HGT	Height Above Surface Level	>= 46 m HGT	D-1653
	LMC	Landmark Category	OR < 46 m HGT	R-2248
	ZVL	Z Value	and LMC 1	L-3801
				O-3156
				R-0046
1F070 FLARE PIPE				
Point	HGT	Height Above Surface Level	LOC 2	D-1653
	LOC	Location /Origin Category	OR LOC 3	R-2248
	ZVL	Z Value	and >= 46 m HGT	R-2251
				L-3801
				O-3156
				R-0046

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: INSTITUTIONAL /GOVERNMENTAL (1H)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1H050	FORT			
Area	LMC	Landmark Category		L-3801
	NAM	Name Category		L-4008
	WID	Width	> 40 m WID	L-4813
Point	LMC	Landmark Category		L-4008
	NAM	Name Category		D-1653
	WID	Width	<= 40 m WID	L-3801
				O-3156
				L-4813

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: RESIDENTIAL (1I)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1I020	MOBILE HOME PARK			
	Area	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	ARA	Landmark Category	OR LMC 1	O-3156
	LMC			R-2494
				R-3730
				R-3732
				R-3733

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: AGRICULTURAL (1J)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1J030	FEED LOT /STOCKYARD /HOLDING PEN			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801 R-2243 R-3730 R-3732 R-3733
	LMC	Landmark Category		
1J050	WINDMILL /WINDMOTOR			
Point	HGT	Height Above Surface Level	< 46 m HGT	R-0046
	LMC	Landmark Category	and LMC 1	O-3156
	ZVL	Z Value	OR >= 46 m HGT	D-1653 R-2248 L-3801

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: RECREATIONAL (1K)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1K020	AMUSEMENT PARK ATTRACTION			
Point	APS	Amusement Park Structure		D-1653
	HGT	Height Above Surface Level	>= 46 m HGT	L-3801
	LMC	Landmark Category		R-2248
	ZVL	Z Value		R-0046
1K030	AMUSEMENT PARK			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	LMC	Landmark Category	OR LMC 1	L-4008
	NAM	Name Category		R-2243
				R-2494
				R-3730
				R-3732
				R-3733
				L-4813
1K040	ATHLETIC FIELD			
Area	ARA	Area Coverage Attribute	>= 125 m LEN	L-3801
	LEN	Length /Diameter	and >= 65 m WID	L-4008
	LMC	Landmark Category	OR LMC 1	R-2243
	NAM	Name Category		R-2494
	WID	Width		L-4813
1K060	CAMPGROUND /CAMPSITE			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	LMC	Landmark Category	OR LMC 1	L-4008
	NAM	Name Category		R-2243
				R-2494
				R-3730
				R-3732
				R-3733
				L-4813

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: RECREATIONAL (1K)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1K070	DRIVE-IN THEATER			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	LMC	Landmark Category	OR LMC 1	R-2243
				R-2494
				R-3730
				R-3732
				R-3733
1K090	FAIRGROUNDS			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	LMC	Landmark Category		L-4008
	NAM	Name Category		R-2243
				R-2494
				R-3730
				R-3732
				R-3733
				L-4813
1K100	GOLF COURSE			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	LMC	Landmark Category		L-4008
	NAM	Name Category		R-2243
				R-2494
				R-3730
				R-3732
				R-3733
				L-4813

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: RECREATIONAL (1K)

Fcode	Feature Acode	Attribute	Inclusion Condition	Rule
1K115	OUTDOOR THEATRE /AMPITHEATRE			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	LMC	Landmark Category	OR LMC 1	L-4008
	NAM	Name Category		R-2243
				R-2494
				R-3730
				R-3732
				R-3733
				L-4813
1K130	RACE TRACK			
Line	LEN	Length /Diameter	>= 65 m LEN	L-3801
	LMC	Landmark Category	OR LMC 1	L-4008
	NAM	Name Category		L-4260
				L-4813
1K150	SKI JUMP			
Line	HGT	Height Above Surface Level	>= 125 m LEN	L-3801
	LEN	Length /Diameter	and < 46 m HGT	
Point	HGT	Height Above Surface Level	< 125 m LEN	D-1653
	LEN	Length /Diameter	OR >= 125 m LEN	L-3801
	ZVL	Z Value	and >= 46 m HGT	R-0046
1K160	STADIUM			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	HGT	Height Above Surface Level	OR LMC 1	L-4008
	LMC	Landmark Category		R-2240
	NAM	Name Category		R-2494
				R-3730
				R-3732
				R-3733
				L-4813

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: RECREATIONAL (1K)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1K170	SWIMMING POOL			
Area	ARA	Area Coverage Attribute	>= 125 m LEN	L-3801
	LEN	Length /Diameter	OR >= 65 m WID	R-2243
	LMC	Landmark Category	OR LMC 1	O-3155
	WID	Width		R-9037
1K180	ZOO			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	LMC	Landmark Category	OR LMC 1	L-4008
	NAM	Name Category		R-2243
				R-2494
				R-3730
				R-3732
				R-3733
				L-4813

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: MISCELLANEOUS FEATURES (1L)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1L015	BUILDING			
Area	ARA	Area Coverage Attribute	> 25 m WID and not enclosed by 1L020 OR enclosed by 1L020 and ARA >= 15625 m sq. OR > 25 m WID and enclosed by 1L020 and LMC 1	L-4018 D-1652 L-3801 L-3959 L-3960 L-4008 L-4027 L-4028 O-3156 R-2252 R-2265 R-2337 R-2340 R-2341 R-2491 R-2495
	BFC	Building Function Category		
	EXS	Existence Category		
	HGT	Height Above Surface Level		
	HWT	House of Worship Type		
	LEN	Length /Diameter		
	LMC	Landmark Category		
	NAM	Name Category		
	PRO	Product Category		
	WID	Width		
Line	BFC	Building Function Category	> 25 m LEN and <= 25 m WID and not enclosed by 1L020 OR enclosed by 1L020 and >25m LEN and <= 25m WID and LMC 001	D-1652 L-3959 L-3960 L-4008 L-4027 L-4028 R-2252 R-2265 R-2337 R-2340 R-2341 R-2491 R-2495 L-4018
	EXS	Existence Category		
	HGT	Height Above Surface Level		
	HWT	House of Worship Type		
	LEN	Length /Diameter		
	LMC	Landmark Category		
	NAM	Name Category		
	PRO	Product Category		
	WID	Width		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: MISCELLANEOUS FEATURES (1L)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1L015	BUILDING (Continued)			
Point	BFC	Building Function Category	LEN >= 3m	L-4018
	EXS	Existence Category	and <25m LEN	R-3740
	HGT	Height Above Surface Level	and not enclosed by 1L020	L-3959
	HWT	House of Worship Type	OR HGT >= 46m.	L-3960
	LEN	Length /Diameter	OR LMC 001	L-4008
	LMC	Landmark Category		R-2337
	NAM	Name Category		R-0046
	PRO	Product Category		R-2340
				R-2341
				R-2252
				R-2265
				R-2491
				L-4027
				L-4028
				R-2495
				O-3156
				L-3801
				D-1652
				L-4813
1L020	BUILT-UP AREA			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-3729
	BAC	Built-Up Area Classification		L-3801
	EXS	Existence Category		L-4019
	NAM	Name Category		L-4020
	PPL	Populated Place Category		L-4021
				L-4022
				L-4023
				L-4024
				L-4025
				L-4026
				R-2305
				R-2318
				R-2333
				R-2334
				R-2345
				R-2488
				L-1650

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: MISCELLANEOUS FEATURES (1L)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1L025	CAIRN			
Point	LMC	Landmark Category	LMC 1	N-0000
1L030	CEMETERY			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-4008
	LMC	Landmark Category		R-2243
	NAM	Name Category		L-3801
				R-3730
				R-3732
				R-3733
				L-4813
Point	ARA	Area Coverage Attribute	< 15,625 m square ARA	D-1653
	COC	Conspicuous Object Category	and (REL 0 or 1 or 2 or 3 or 4)	O-3156
	EXS	Existence Category	and LMC 1	
	LMC	Landmark Category	OR EXS 31	
	REL	Religious Denomination	and < 15,625 m square ARA	
			and (REL 0 or 1 or 2 or 3 or 4)	
			OR COC 1	
			and < 15,625 m square ARA	
			and (REL 0 or 1 or 2 or 3 or 4)	
1L060	DRAGON (TIGER) TEETH			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-3730
	WID	Width	and >= 125 m WID	R-3732
				R-3733
				O-3156
Line	WID	Width	< 125 m WID	O-3156
1L070	FENCE			
Line	LEN	Length /Diameter	>= 1.5 m PFH	R-2352
	LMC	Landmark Category	and >= 2,500 m LEN	R-2353
	PFH	Predominant Feature Height	OR LMC 1	

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: MISCELLANEOUS FEATURES (1L)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1L085	GEOPHYSICAL PROSPECTING GRID			
Line	LEN	Length /Diameter	>= 625 m LEN	N-0000
1L100	HUT			
Point	LMC	Landmark Category	LMC 1	R-2343
1L130	MONUMENT			
Point	COC	Conspicuous Object Category	>= 46 m HGT	L-3801
	HGT	Height Above Surface Level	OR < 46 m HGT	L-4008
	LMC	Landmark Category	and LMC 1	R-2248
	NAM	Name Category	OR COC 1	R-0046
	SSC	Structure Shape Category		O-3156
	ZVL	Z Value		
1L140	NUCLEAR ACCELERATOR			
Area	ARA	Area Coverage Attribute	> 40 m LEN	L-3801
	LEN	Length /Diameter		R-2243
	LMC	Landmark Category		O-3156

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: MISCELLANEOUS FEATURES (1L)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1L160	PIPELINE /PIPE			
Line	ACC	Accuracy Category	LMC 1	L-4260
	DEP	Depth Below Surface Level	OR > 1,250 m LEN	R-3705
	EXS	Existence Category	and (LOC 1 or 3 or 4)	L-3801
	HSB	Height Above Sea Bottom	and (PRO 0 or 6 or 12 or 13 or 18 or 27)	L-4006
	LEN	Length /Diameter		L-4012
	LMC	Landmark Category	OR (LOC 10 or 11 or 12)	L-4013
	LOC	Location /Origin Category	OR (LOC 1 or 3 or 4)	L-4014
	OWO	Over Water Obstruction	and OWO 1	L-4743
	PLT	Pipeline Type		L-4862
	PRO	Product Category		O-3427
				R-2208
				R-2214
				R-2249
				R-2348
				R-2349
				R-2818
				R-2937
				O-3156
1L170	PLAZA /CITY SQUARE			
Area	NAM	Name Category	If WID is > WID of entering Roads (1P030).	L-4008
	WID	Width		R-2313
1L180	PUMPING STATION			
Area	HGT	Height Above Surface Level	>= 125 m WID	L-3801
	LMC	Landmark Category	and (PRO 12 or 18 or 19 or 27)	R-2240
	PRO	Product Category		
	WID	Width		
Point	HGT	Height Above Surface Level	< 125 m WID	L-3801
	LMC	Landmark Category	and (PRO 12 or 18 or 19 or 27)	R-2240
	PRO	Product Category		O-3156
	WID	Width		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: MISCELLANEOUS FEATURES (1L)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1L200	RUINS			
Area	ARA	Area Coverage Attribute	LOC 2	L-3801
	HDI	Hydrographic Depth /Height Information	and > 200 m LEN OR LOC 3	L-4008
	HDP	Hydrographic Depth	and > 40 m LEN	L-4702
	HGT	Height Above Surface Level		L-4722
	LEN	Length /Diameter		L-4729
	LMC	Landmark Category		L-4807
	LOC	Location /Origin Category		R-2221
	NAM	Name Category		R-2222
	VDC	Vertical Datum Category		R-2240
	VDR	Vertical Datum Record		R-2243
	VRC	Vertical Reference Category		R-2800
				R-2806
				R-3672
				R-3708
Point	HDI	Hydrographic Depth /Height Information	LOC 2	D-1909
	HDP	Hydrographic Depth	and <= 200 m LEN OR LOC 3	L-3801
	HGT	Height Above Surface Level	and <= 40 m LEN	L-4702
	LEN	Length /Diameter	and LMC 1	L-4722
	LMC	Landmark Category		L-4729
	LOC	Location /Origin Category		R-2221
	VDC	Vertical Datum Category		R-2222
	VDR	Vertical Datum Record		R-2806
	VRC	Vertical Reference Category		R-3672
				R-3708
				R-3709
1L208	SHANTYTOWN			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-3730
				R-3732
				R-3733
				O-3156
1L210	SNOW SHED /ROCK SHED			
Line	LEN	Length /Diameter	>= 40 m LEN	R-2254
	SIT	Shed Identifier Type		L-3801

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: MISCELLANEOUS FEATURES (1L)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1L210	SNOW SHED /ROCK SHED (Continued)			
Point	LEN	Length /Diameter	< 40 m LEN	R-2253 R-2254 L-3801 O-3156
	SIT	Shed Identifier Type		
1L228	TENT DWELLINGS			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-4008 R-3730 R-3732 R-3733
	NAM	Name Category		
Point	ARA	Area Coverage Attribute	< 15,625 m square ARA and LMC 1	L-4008
	LMC	Landmark Category		
	NAM	Name Category		
1L240	TOWER (NON- COMMUNICATION)			
Point	HGT	Height Above Surface Level	All required	L-3801 R-2240 R-0046
	LMC	Landmark Category		
	ZVL	Z Value		
1L250	UNDERGROUND DWELLING			
Point	LMC	Landmark Category	LMC 1	N-0000
1L260	WALL			
Line	LEN	Length /Diameter	> 1.5 m PFH and > 125 m LEN OR LMC 1	R-2353 R-2250 L-3801 O-3156
	LMC	Landmark Category		
	PFH	Predominant Feature Height		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: STORAGE (1M)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1M010	**DEPOT (STORAGE)**			
Area	LMC	Landmark Category	>= 125 m WID	O-3156
	LOC	Location /Origin Category	OR LMC 1	L-4016
	WID	Width		R-2494
				L-3801
1M020	**GRAIN BIN**			
Area	LEN	Length /Diameter	>= 40 m LEN	L-3801
Point	LEN	Length /Diameter	< 40 m LEN	L-3801
				O-3156
1M030	**GRAIN ELEVATOR**			
Area	HGT	Height Above Surface Level	>= 40 m LEN	L-3801
	LEN	Length /Diameter	and < 46 m HGT	
Point	HGT	Height Above Surface Level	>= 46 m HGT	L-3801
	LEN	Length /Diameter	OR < 46 m HGT	R-0046
	LMC	Landmark Category	and LMC 1	O-3156
	ZVL	Z Value	and < 40 m LEN	
1M050	**SILO**			
Point	HGT	Height Above Surface Level	>= 46 m HGT	L-3801
	LMC	Landmark Category	OR < 46 m HGT	R-0046
	ZVL	Z Value	and LMC 1	O-3156
1M060	**STORAGE BUNKER /STORAGE MOUND**			
Area	LEN	Length /Diameter	>= 40 m LEN	L-4006
	PRO	Product Category		L-3801
Point	LEN	Length /Diameter	< 40 m LEN	L-4006
	PRO	Product Category		O-3156
				L-3801

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: STORAGE (1M)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1M070	TANK			
Area	HGT	Height Above Surface Level	> 40 m LEN	L-3801
	LEN	Length /Diameter	and < 46 m HGT	L-4006
	LOC	Location /Origin Category		L-4029
	PRO	Product Category		T-0301
Point	HGT	Height Above Surface Level	<= 40 m LEN	L-3801
	LEN	Length /Diameter	and < 46 m HGT	L-4006
	LOC	Location /Origin Category	OR >= 46 m HGT	L-4029
	PRO	Product Category	and LOC 3	R-0046
	ZVL	Z Value		O-3156
				T-0301
1M080	WATER TOWER			
Point	HGT	Height Above Surface Level	All required	L-3801
	LMC	Landmark Category		R-2240
	ZVL	Z Value		R-0046
				O-3156

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: TRANSPORTATION R/R (1N)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1N010	RAILROAD TRACK			
Line	ACC	Accuracy Category	All required	D-1650
	EXS	Existence Category		L-3801
	GAW	Gauge Width		L-3956
	LOC	Location /Origin Category		L-3957
	LTN	Lane /Track Number		L-3961
	NAM	Name Category		L-3962
	RGC	Railroad Gauge Category		L-3963
	RPS	Railroad Power Source		L-4008
	RRC	Railroad /Road Categories		R-2229
	VRC	Vertical Reference Category		R-2315
				R-2324
				R-2327
				R-2328
				R-2329
				R-2330
				R-3706
				L-4260
				R-3801
				R-3672
1N050	RR SIDING /RR SPUR			
Line	EXS	Existence Category	All required	D-1651
	RGC	Railroad Gauge Category		R-2326
	RPS	Railroad Power Source		R-2239
	RSA	Rail Siding /Spur Attribute		L-3801
1N075	RR TURNTABLE			
Point			All required	N-0000
1N080	RR YARD			
Area	EXS	Existence Category	All required	L-3562
				L-3801
				R-2238

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: TRANSPORTATION R/R (1N)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1N090	TRAMWAY /INCLINE RAILWAY			0-3156
Line	LEN	Length /Diameter	>= 375 m LEN	
	LMC	Landmark Category	OR LMC 1	

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: TRANSPORTATION /ROADS (1P)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1P010	CART TRACK			
Line	WTC	Route Weatherability Category	(WTC 2 or 3)	R-2341 O-3156
1P020	INTERCHANGE			
Line	EXS	Existence Category	RST 1	R-2233
	LOC	Location /Origin Category	and (TUC 4 or 7)	O-3156
	LTC	Lane /Track Characteristics	and >= 2.5 m WTW	R-2234
	LTN	Lane /Track Number	and (EXS 5 or 28)	
	RST	Road /Runway Surface Type	and (LOC 3 or 4)	
	TUC	Transportation Use Category	and WTC 1	
	WTC	Route Weatherability Category		
	WTW	Width of Travelled Way		
1P030	ROAD			
Line	ACC	Accuracy Category	>= 2.5 m WTW	L-4260
	EXS	Existence Category	and (EXS 5 or 28)	L-3953
	LOC	Location /Origin Category	and (LOC 3 or 4)	L-3801
	LTC	Lane /Track Characteristics		O-3156
	LTN	Lane /Track Number		L-3952
	MED	Median Category		L-3955
	NAM	Name Category		L-3951
	RST	Road /Runway Surface Type		L-4008
	TUC	Transportation Use Category		L-4034
	WTC	Route Weatherability Category		O-3150
	WTW	Width of Travelled Way		R-2233 R-2234 R-2300 R-2301 R-2303 R-2304 R-2305 R-2315 R-2330

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: TRANSPORTATION /ROADS (1P)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1P050	TRAIL			
	Line	Landmark Category	(WTC 0 or 2 or 3)	L-4033
	LMC	Route Weatherability Category	and LMC 1	O-3156
	WTC			

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: ASSOCIATED TRANSPORTATION (1Q)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1Q010	**AERIAL CABLEWAY LINE /SKI LIFT LINE**			
Line	LEN	Length /Diameter	>= 375 m LEN	L-3801
	LMC	Landmark Category	OR LMC 1	
	OWO	Over Water Obstruction	OR OWO 1	
	USE	Use Status		
1Q040	**BRIDGE /OVERPASS /VIADUCT**			
Line	BOT	Bridge Opening Type	(TUC 1 or 3 or 4)	L-4260
	LEN	Length /Diameter	and >= 75 m LEN	R-2236
	LMC	Landmark Category	OR TUC 17	R-2315
	TUC	Transportation Use Category	and LMC 1	R-2316
			and >= 75 m LEN	R-9035
Point	LEN	Length /Diameter	(TUC 1 or 3 or 4)	O-3156
	LMC	Landmark Category	and < 75 m LEN	
	TUC	Transportation Use Category	OR TUC 17	
			and LMC 1	
			and < 75 m LEN	
1Q060	**CONTROL TOWER**			
Point	HGT	Height Above Surface Level	All required	R-2495
	ZVL	Z Value		L-3801
				R-0046
1Q065	**CULVERT**			
Point	LEN	Length /Diameter	>= 5 m LEN	N-0000
	WGP	Width with Greater Precision	and >= 1.5 m WGP	

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: ASSOCIATED TRANSPORTATION (1Q)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1Q070	FERRY CROSSING			
Line	EXS	Existence Category	>= 25 m FCL	R-2320
	FCL	Ferry Crossing Length	and (TUC 1 or 3 or 4 or 17)	L-4008
	FER	Ferry Type	and EXS 28	L-4032
	LMC	Landmark Category		R-2232
	NAM	Name Category		L-4813
	TUC	Transportation Use Category		
Point	EXS	Existence Category	< 25 m FCL	L-4031
	FCL	Ferry Crossing Length	and (TUC 1 or 3 or 4)	L-4032
	LMC	Landmark Category	and EXS 28	L-4008
	NAM	Name Category	OR TUC 17	R-2232
	TUC	Transportation Use Category	and < 25 m FCL	L-4813
			and EXS 28	
			and LMC 1	
1Q080	FERRY SITE /FERRY SLIP			
Point	LEN	Length /Diameter	(TUC 1 or 4)	L-4779
	TUC	Transportation Use Category		
	WID	Width		
1Q100	DISTANCE MARKER			
Point	DVA	Distance Value Attribute	UNI 10	L-3801
	UNI	Units Category		L-3606
				R-3736
				L-4878
1Q110	MOORING MAST			
Point	HGT	Height Above Surface Level	All required	L-3801
	ZVL	Z Value		R-0046

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: ASSOCIATED TRANSPORTATION (1Q)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1Q115	REST AREA /VEHICLE STOPPING AREA			
Area	LMC	Landmark Category	>= 125 m WID	R-2494
	WID	Width	and LMC 1	R-2231
				L-3801
1Q116	ROUTE MARKER			
Point	NAM	Name Category	All required in association with	L-3801
	USE	Use Status	1P030 (USE 4 or 5 or 23)	L-3996
				R-2260
				R-2264
				R-2302
				R-2307
				R-2312
				R-3692
1Q131	TUNNEL			
Line	LEN	Length /Diameter	>= 75 m LEN	L-4260
	NAM	Name Category		L-4008
	TRA	Traversability Attribute		R-2263
	TUC	Transportation Use Category		R-2318
				R-2325
				L-4813
1Q140	VEHICLE STORAGE /VEHICLE PARKING			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-2494
	LMC	Landmark Category	and MOT 4	R-3730
	MOT	Mode of Transport	OR LMC 1	R-3732
				R-3733

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: COMMUNICATION /TRANSMISSION (1T)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1T005	CABLE			
Line	EXS	Existence Category	LOC 6	R-2211
	LOC	Location /Origin Category		R-2212
	USE	Use Status		R-2818
1T010	DISH			
Point	HGT	Height Above Surface Level	LMC 1	L-3801
	LMC	Landmark Category	and < 46 m HGT	O-3156
	ZVL	Z Value	OR >= 46 m HGT	R-0046
1T030	POWER TRANSMISSION LINE			
Line	ACC	Accuracy Category	>= 375 m LEN	L-3801
	LEN	Length /Diameter	OR OWO 1	O-3156
	LMC	Landmark Category		R-2492
	OWO	Over Water Obstruction		L-4260
				R-2275
1T050	STATION (COMMUNICATION)			
Area	LEN	Length /Diameter	(NST 12 or 13 or 15 or 16 or 33)	L-4008
	NAM	Name Category	and > 40 m LEN	L-3801
	NST	Radio Navigation /Communication		L-4813
Point	LEN	Length /Diameter	(NST 12 or 13 or 15 or 16 or 33)	L-4008
	NAM	Name Category	and <= 40 m LEN	O-3156
	NST	Radio Navigation /Communication		L-3801
				L-4813
1T060	TELEPHONE LINE /TELEGRAPH LINE			
Line	EXS	Existence Category	>= 2,500 m LEN	R-2351
	LEN	Length /Diameter	and LMC 1	L-3801
	LMC	Landmark Category	OR OWO 1	O-3156
	OWO	Over Water Obstruction		L-4260

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: COMMUNICATION /TRANSMISSION (1T)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1T080	TOWER (COMMUNICATION)			
	Point HGT	Height Above Surface Level	(NST 12 or 15 or 16 or 33)	O-3156
	LMC	Landmark Category	and >= 46 m HGT	R-2240
	NAM	Name Category	OR (NST 12 or 15 or 16 or 33)	L-3801
	NST	Radio Navigation /Communication	and < 46 m HGT	L-4813
	ZVL	Z Value		

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TABLE I Feature/Attribute category, inclusion conditions and product generation rules. - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: AIRPORTS (1U)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1U030	AIRCRAFT FACILITY			
Area	AFT	Aircraft Facility Type	(AFT 1 or 2 or 3)	L-5011
	ARA	Area Coverage Attribute	and (USE 0 or 8 or 22 or 49)	L-3801
	COD	Certainty of Delineation	and COD 1	L-4008
	EXS	Existence Category	and (EXS 6 or 28)	R-2356
	LEN	Length /Diameter	and >= 125 m LEN	R-2494
	NAM	Name Category	and >= 50 m WID	R-2495
	USE	Use Status		L-4813
	WID	Width		
	ZVL	Z Value		
Point	AFT	Aircraft Facility Type	AFT 1	O-3156
	COD	Certainty of Delineation	and (USE 0 or 8 or 22 or 49)	L-5011
	EXS	Existence Category	and (EXS 6 or 28)	L-3801
	LEN	Length /Diameter	and COD 2	L-4008
	NAM	Name Category	OR (AFT 2 or 3)	R-2356
	USE	Use Status	and (USE 0 or 8 or 22 or 49)	
	WID	Width	and (EXS 6 or 28)	
	ZVL	Z Value	and < 125 m LEN	
			OR (AFT 2 or 3)	
			and (USE 0 or 8 or 22 or 49)	
			and (EXS 6 or 28)	
			and < 50 m WID	
			OR (AFT 2 or 3)	
			and (USE 0 or 8 or 22 or 49)	
			and (EXS 6 or 28)	
			and COD 2	
1U040	AIRCRAFT FACILITY BEACON			
Point	COL	Character of Light	All required	R-2849
1U060	APRON /HARDSTAND			
Area	WID	Width	>= 20 m WID	N-0000
1U130	OVERRUN /STOPWAY			
Area			All required	N-0000

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : CULTURE (1)
SUBCATEGORY: AIRPORTS (1U)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
1U130	OVERRUN /STOPWAY			
Area			All required	N-0000
1U160	RUNWAY			
Area	EXS	Existence Category	All required	L-4017
	RST	Road /Runway Surface Type		L-3801
	WID	Width		O-3156
	ZVL	Z Value		
1U190	SEAPLANE LANDING OR TAKE-OFF AREA			
Area	EXS	Existence Category	EXS 1	L-4705 L-4722 R-2986
1U200	TAXIWAY			
Area	EXS	Existence Category	>= 350 m LEN	O-3156
	LEN	Length /Diameter	and (EXS 0 or 28)	

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: COASTAL HYDRO (2A)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2A010	COASTAL SHORELINE			
Line	ACC	Accuracy Category	All required	R-2024
	SLT	Shoreline Type Category		R-2023
	VDC	Vertical Datum Category		L-4132
	VDR	Vertical Datum Record		R-3735
				R-2316
				R-2372
				R-2437
				R-2440
				D-7010
				R-1200
2A020	FORESHORE			
Area	LEN	Length /Diameter	>= 150 m LEN	L-4705
	MCP	Material Composition Primary		L-4706
	MCS	Material Composition Secondary		R-2202
				R-2316
				R-2825
				R-2826
				L-4722
Point	LEN	Length /Diameter	< 150 m LEN	L-4706
	MCP	Material Composition Primary		L-4722
	MCS	Material Composition Secondary		R-2202
				R-2825
				R-2826
				R-3709
2A040	OPEN WATER (EXCEPT INLAND)			
Area			All required	N-0000

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TABLE I Feature/Attribute category, inclusion conditions and product generation rules.- Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: PORTS AND HARBORS (2B)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2B010	ANCHORAGE			
Area	ANC	Anchorage Type Category	COD = 1	R-2800
	COD	Certainty of Delineation	and LEN > 400 m	R-2901
	LEN	Length /Diameter		R-2811
	NAM	Name Category		L-4705
	TIM	Time Attribute		L-4714
	WID	Width		L-4796
				L-4797
				L-4866
				L-4869
				L-4722
				L-4813
Point	ANC	Anchorage Type Category	COD = 1	L-4869
	COD	Certainty of Delineation	and LEN <= 400 m	L-4866
	LEN	Length /Diameter	OR COD = 2	
	NAM	Name Category		
	TIM	Time Attribute		
	WID	Width		
2B040	BREAKWATER			
Area	LEN	Length /Diameter	> 20 m WID	L-4725
	VRC	Vertical Reference Category		R-2202
	WID	Width		R-2802
				R-2803
				R-3672
				R-3700
Line	LEN	Length /Diameter	<= 20 m WID	L-4725
	VRC	Vertical Reference Category		L-4743
	WID	Width		R-3700
2B080	DOLPHIN			
Point			All required	L-4722
				L-4737
				L-4800

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: PORTS AND HARBORS (2B)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2B090	DRY DOCK			
Area	LEN	Length /Diameter	LOC 7	R-2804
	LOC	Location /Origin Category	and >= 20 m WID	R-2904
	NAM	Name Category	OR LOC 5	R-3675
	WID	Width		L-4705
				L-4709
				L-4722
2B100	FISHERY /FISH STAKES			R-2905
Line			All required	
2B110	FISH TRAP /FISH WEIR			L-4722
Area			All required	D-7011
2B115	GRIDIRON			N-0000
Area	LEN	Length /Diameter	>= 200 m LEN	
2B140	JETTY			R-2202
Area	LEN	Length /Diameter	> 20 m WID	R-2802
	VRC	Vertical Reference Category		R-2803
	WID	Width		R-3672
				R-3700
Line	LEN	Length /Diameter	<= 20 m WID	R-3700
	VRC	Vertical Reference Category		
	WID	Width		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: PORTS AND HARBORS (2B)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2B150	LANDING PLACE			
Area	COD	Certainty of Delineation	HOC 5	L-4802
	HOC	Hydrographic Origin Category	and COD 1	
	LEN	Length /Diameter	and > 150 m LEN	
Point	COD	Certainty of Delineation	HOC 5	L-4802
	HOC	Hydrographic Origin Category	and COD 1	R-3668
	LEN	Length /Diameter	and <= 150 m LEN	
			OR HOC 5	
			and COD 2	
			OR HOC 4	
2B155	MARITIME STATION			
Point	SST	Sound Signal Type	STN 17	L-4722
	STN	Maritime Station Type	and (SST 0 or 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8)	
2B170	OFFSHORE LOADING FACILITY			
Area	LEN	Length /Diameter	>= 40 m WID	L-4709
	NAM	Name Category	and >= 40 m LEN	L-4860
	WID	Width		L-4705
				R-9035
				L-4722
Line	LEN	Length /Diameter	< 40 m WID	L-4709
	NAM	Name Category	and >= 40 m LEN	L-4860
	WID	Width		L-4743
				R-9035
Point	CHA	Light Characteristic Category	< 40 m WID	R-2849
	LEN	Length /Diameter	and < 40 m LEN	L-4709
	NAM	Name Category		L-4722
	USE	Use Status		L-4860
	WID	Width		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: PORTS AND HARBORS (2B)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2B190	PIER, WHARF			
Area	PUC	Pier Use Category		R-3700
	WID	Width	>= 20 m WID	R-2804
				R-9035
Line	PUC	Pier Use Category		R-3700
	WID	Width	< 20 m WID	
2B220	RAMP			
Area	VRC	Vertical Reference Category		L-4803
	WID	Width	> 20 m WID	R-2202
				R-2802
				R-2803
				R-3672
Line	VRC	Vertical Reference Category		L-4803
	WID	Width	<= 20 m WID	
2B230	SEAWALL			
Line			All required	R-3700
2B240	SLIPWAY			
Line	VRC	Vertical Reference Category		L-4803
			All required	R-2202
				R-2803
				R-3672

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: NAVAIDS (2C)

Fcode	Feature Acode	Attribute	Inclusion Condition	Rule
2C010	BUOY			
Point	BF1	Broadcast Frequency (1)	All required	L-4868
	BF2	Broadcast Frequency (2)		R-2832
	CCF	Color Code of Feature		L-4876
	COL	Character of Light		L-4875
	EOL	Elevation of Light		L-4709
	LVR	Light Visibility Range		L-4711
	MLR	Multiple Light Ranges		L-4737
	NAM	Name Category		S-1403
	PER	Period of Light		L-4759
	RA1	Radio Aid (1)		L-4761
	RA2	Radio Aid (2)		L-4766
	REF	Radar Reflector Attribute		L-4767
	SSC	Structure Shape Category		L-4768
	SST	Sound Signal Type		L-4789
	TMC	Topmark Category		L-4790
				L-4831
				L-4832
				L-4833
				L-4834
				L-4835
				L-4836
				L-4837
				L-4838
				L-4839
				L-4840
				L-4841
				L-4842
				L-4843
				L-4844
				L-4845
				L-4846
				L-4849
				L-4850
				L-4853
				R-2717
				R-2718
				R-2719
				R-2720
				R-2721
				R-2722
				R-2723

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
PRODUCT : COMBAT				
CATEGORY : HYDROGRAPHY (2)				
SUBCATEGORY: NAVAIDS (2C)				
2C010	BUOY (Continued)			R-2724
				R-2725
				R-2726
				R-2727
				R-2849
				R-2884
				R-2885
				R-2886
				R-2887
				R-2920
				R-2992
				R-2994
				R-2995
				R-2996
				R-2997
				R-3684
				D-7013
Point				
2C020	CLEARING LINE			R-2999
				O-3420
				L-4743
				L-4830
				L-7010
				D-7012
Line	BRG	Bearing of Object	(LAF 1 or 2 or 3 or 4 or 5)	
	DRP	Description of Reference Point		
	LAF	Line Associated Features		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: NAVAIDS (2C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2C030	ELECTRONIC BEACON			
Point	BF1	Broadcast Frequency (1)	All required	O-3400
	BF2	Broadcast Frequency (2)		L-4709
	BR1	Beacon Range (1)		L-4737
	BR2	Beacon Range (2)		L-4783
	NAM	Name Category		L-4832
	RA1	Radio Aid (1)		L-4835
	RA2	Radio Aid (2)		L-4836
				L-4840
				L-4841
				L-4844
				L-4850
				L-4853
				T-0854
				T-0855
				D-7013
2C040	LEADING LINE			
Line	BRG	Bearing of Object	All required	R-2728
	DRP	Description of Reference Point		R-2998
	LAF	Line Associated Features		R-3681
				O-3420
				L-4743
				L-4855
				L-7010
				D-7012

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: NAVAIDS (2C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2C050	LIGHT			
	Point BF1	Broadcast Frequency (1)	(HLT 0 or 1 or 2 or 3)	S-1402
	BF2	Broadcast Frequency (2)		R-2716
	CCF	Color Code of Feature		R-2729
	COL	Character of Light		R-2849
	EOL	Elevation of Light		R-2884
	HLT	Hydrographic Light Type		R-2887
	IAC	IALA Aid Category		R-2889
	L51	Sector Label (1)		R-2920
	L52	Sector Label (2)		R-2992
	L53	Sector Label (3)		O-3400
	L54	Sector Label (4)		O-3415
	L55	Sector Label (5)		L-4709
	L56	Sector Label (6)		L-4711
	L57	Sector Label (7)		L-4737
	L58	Sector Label (8)		L-4759
	L59	Sector Label (9)		L-4760
	L60	Sector Label (10)		L-4761
	L61	Sector Label (11)		L-4762
	L62	Sector Label (12)		L-4788
	L63	Sector Label (13)		L-4789
	L64	Sector Label (14)		L-4790
	L65	Sector Label (15)		L-4792
	L66	Sector Label (16)		L-4793
	L67	Sector Label (17)		L-4783
	L68	Sector Label (18)		L-4831
	L69	Sector Label (19)		L-4832
	L70	Sector Label (20)		L-4833
	L71	Sector Label (21)		L-4834
	L72	Sector Label (22)		L-4835
	L73	Sector Label (23)		L-4836
	L74	Sector Label (24)		L-4837
	L75	Sector Label (25)		L-4838
	LVR	Light Visibility Range		L-4839
	MLR	Multiple Light Ranges		L-4840
	NAM	Name Category		L-4841
	PER	Period of Light		L-4842
	RA1	Radio Aid (1)		L-4843
	RA2	Radio Aid (2)		L-4844
	REF	Radar Reflector Attribute		L-4845
	S51	Sector Angle (1)		L-4847
	S52	Sector Angle (2)		L-4848

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: NAVAIDS (2C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2C050	LIGHT (Continued)			
Point	S53	Sector Angle (3)		L-4849
	S54	Sector Angle (4)		L-4850
	S55	Sector Angle (5)		L-4851
	S56	Sector Angle (6)		L-4852
	S57	Sector Angle (7)		L-4853
	S58	Sector Angle (8)		L-4856
	S59	Sector Angle (9)		L-4857
	S60	Sector Angle (10)		L-4858
	S61	Sector Angle (11)		R-3681
	S62	Sector Angle (12)		R-3682
	S63	Sector Angle (13)		R-3683
	S64	Sector Angle (14)		T-0853
	S65	Sector Angle (15)		L-4865
	S66	Sector Angle (16)		L-4867
	S67	Sector Angle (17)		R-3685
	S68	Sector Angle (18)		L-4868
	S69	Sector Angle (19)		T-0826
	S70	Sector Angle (20)		R-2832
	S71	Sector Angle (21)		L-4875
	S72	Sector Angle (22)		L-4876
	S73	Sector Angle (23)		D-7013
	S74	Sector Angle (24)		R-2259
	S75	Sector Angle (25)		
	SST	Sound Signal Type		
	TMC	Topmark Category		
2C055	MARKER			
Point			All required	L-4722 D-7013

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: NAVAIDS (2C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2C060	VISUAL BEACON			
Point	BF1	Broadcast Frequency (1)	All required	L-4709
	BF2	Broadcast Frequency (2)		L-4737
	CCF	Color Code of Feature		L-4868
	NAM	Name Category		L-4783
	RA1	Radio Aid (1)		L-4790
	RA2	Radio Aid (2)		L-4793
	REF	Radar Reflector Attribute		L-4831
	SST	Sound Signal Type		L-4832
	TMC	Topmark Category		L-4833
				L-4834
				L-4835
				L-4836
				L-4837
				L-4838
				L-4839
				L-4840
				L-4841
				L-4843
				L-4844
				L-4849
				L-4850
				L-4853
				R-2992
				D-7013

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DANGERS AND UNDERWATER FEATURES (2D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2D000	MISCELLANEOUS UNDERWATER FEATURE			
Area	ACC	Accuracy Category	(SFC 1 or 2 or 3)	L-4700
	DAT	Date Category	and >= 200 m LEN	L-4702
	DDA	Description of Danger		L-4707
	EXS	Existence Category		L-4708
	HDI	Hydrographic Depth /Height Information		L-4722
	HDP	Hydrographic Depth		L-4729
	LEN	Length /Diameter		L-4730
	SFC	Sea Floor Feature Category		L-4807
	SOH	Severity of Hazard		L-4808
	VDC	Vertical Datum Category		O-3411
	VDR	Vertical Datum Record		R-2221
	WID	Width		R-2222
				R-2800
				R-2806
				R-2916
				R-3704
				R-3708
Point	ACC	Accuracy Category	(SFC 4 or 5)	D-1909
	DAT	Date Category	OR (SFC 1 or 2 or 3)	L-4700
	DDA	Description of Danger	and < 200 m LEN	L-4702
	EXS	Existence Category		L-4707
	HDI	Hydrographic Depth /Height Information		L-4708
	HDP	Hydrographic Depth		L-4722
	LEN	Length /Diameter		L-4729
	SFC	Sea Floor Feature Category		L-4730
	SOH	Severity of Hazard		L-4808
	VDC	Vertical Datum Category		L-4872
	VDR	Vertical Datum Record		O-3411
	WID	Width		R-2221
				R-2222
				R-2916
				R-3704
				R-3708
				R-3709
				S-1401

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DANGERS AND UNDERWATER FEATURES (2D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2D010	BREAKERS			
Area	LEN	Length /Diameter	> 250 m WID	R-2910
	WID	Width	and > 750 m LEN	R-2911
Point	ACC	Accuracy Category	<= 250 m WID	L-4700
	DAT	Date Category	OR <= 750 m LEN	L-4707
	EXS	Existence Category		L-4708
	LEN	Length /Diameter		L-4722
	WID	Width		L-4730
				L-4808
				O-3411
				S-1404
2D020	CRIB			
Area	ACC	Accuracy Category	> 125 m LEN	L-4700
	DAT	Date Category		L-4702
	EXS	Existence Category		L-4707
	HDI	Hydrographic Depth /Height Information		L-4708
	HDP	Hydrographic Depth		L-4722
	LEN	Length /Diameter		L-4729
	VDC	Vertical Datum Category		L-4807
	VDR	Vertical Datum Record		L-4808
	VRC	Vertical Reference Category		L-4809
				O-3411
				R-2202
				R-2221
				R-2222
				R-2802
				R-2911
				R-3672
				R-3708

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DANGERS AND UNDERWATER FEATURES (2D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2D020	CRIB (Continued)			
Point	ACC	Accuracy Category	<= 125 m LEN	L-4700
	AOO	Angle of Orientation		L-4702
	DAT	Date Category		L-4707
	EXS	Existence Category		L-4708
	HDI	Hydrographic Depth /Height Information		L-4722
				L-4729
	HDP	Hydrographic Depth		L-4808
	LEN	Length /Diameter		L-4809
	VDC	Vertical Datum Category		O-3411
	VDR	Vertical Datum Record		R-2221
	VRC	Vertical Reference Category		R-2222
				R-2911
				R-3672
				R-3708
2D030	DISCOLORED WATER			
Area	ACC	Accuracy Category	> 200 m LEN	L-4700
	DAT	Date Category		L-4707
	EXS	Existence Category		L-4708
	LEN	Length /Diameter		L-4722
				L-4730
				L-4808
				O-3411
				R-2911
				R-3708
Point	ACC	Accuracy Category	<= 200 m LEN	L-4700
	DAT	Date Category		L-4707
	EXS	Existence Category		L-4708
	LEN	Length /Diameter		L-4722
				L-4730
				L-4808
				O-3411
				R-2911
				R-3708

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DANGERS AND UNDERWATER FEATURES (2D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2D040	EDDIES			
Area	WID	Width	> 250 m WID	R-2912
				R-2913
Point	WID	Width	<= 250 m WID	N-0000
2D050	FOUL GROUND			
Area	ACC	Accuracy Category	> 200 m LEN	L-4700
	DAT	Date Category		L-4702
	EXS	Existence Category		L-4707
	HDI	Hydrographic Depth /Height Information		L-4708
	HDP	Hydrographic Depth		L-4722
	LEN	Length /Diameter		L-4729
	VDC	Vertical Datum Category		L-4730
	VDR	Vertical Datum Record		L-4807
				L-4808
				O-3411
				R-2221
				R-2222
				R-2800
Point	ACC	Accuracy Category	<= 200 m LEN	D-1909
	DAT	Date Category		L-4700
	EXS	Existence Category		L-4702
	HDI	Hydrographic Depth /Height Information		L-4707
	HDP	Hydrographic Depth		L-4708
	LEN	Length /Diameter		L-4722
	VDC	Vertical Datum Category		L-4729
	VDR	Vertical Datum Record		L-4730
				L-4808
				L-4872
				O-3411
				R-2221
				R-2222
				R-2806
				R-3709

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DANGERS AND UNDERWATER FEATURES (2D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2D060	KELP			
Area	WID	Width	> 250 m WID	R-2912 R-2913
Point	WID	Width	<= 250 m WID	N-0000
2D080	OVERFALLS /TIDE RIPS			
Area	NAM	Name Category	> 250 m WID	L-4709 L-4737 R-2912 R-2913
	WID	Width		
Point	WID	Width	<= 250 m WID	N-0000
2D090	PERCH /STAKE			
Area			Area if shown on cartographic source as area	L-4722 L-4729 L-4807 L-4808 L-4809 R-2203 R-2800 R-2914 R-3708
Point			Point if shown as point on cartographic source.	L-4722 R-2203 R-2806 R-2914

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DANGERS AND UNDERWATER FEATURES (2D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2D100	PILING			
Area	ACC	Accuracy Category	Area if shown on cartographic source as area	L-4700
	DAT	Date Category		L-4707
	EXS	Existence Category		L-4708
	VRC	Vertical Reference Category		L-4722
				L-4730
				L-4807
				L-4808
				O-3411
				R-2203
				R-2800
				R-2914
				R-3708
Point	ACC	Accuracy Category	Point if shown as point on cartographic source.	L-4700
	DAT	Date Category		L-4707
	EXS	Existence Category		L-4708
	VRC	Vertical Reference Category		L-4722
				L-4730
				L-4808
				L-4809
				O-3411
				R-2203
				R-2914
2D110	PLATFORM			
Point	CHA	Light Characteristic Category	All required	L-4706
	NAM	Name Category		L-4730
	NST	Radio Navigation /Communication		T-0800
				L-4722
	SST	Sound Signal Type		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DANGERS AND UNDERWATER FEATURES (2D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2D120	REEF			
Area	ACC	Accuracy Category	All required	D-1910
	COD	Certainty of Delineation		L-4700
	DAT	Date Category		L-4702
	EXS	Existence Category		L-4707
	HDH	Hydrographic Drying Height		L-4708
	HDI	Hydrographic Depth /Height Information		L-4722
				L-4730
	HDP	Hydrographic Depth		L-4807
	MCP	Material Composition Primary		L-4808
	NAM	Name Category		L-4809
	SOH	Severity of Hazard		L-4811
	VDC	Vertical Datum Category		O-3411
	VDR	Vertical Datum Record		R-2202
	VRC	Vertical Reference Category		R-2210
				R-2215
				R-2221
				R-2222
				R-2802
				R-2806
				R-2915
				R-3708
				L-4709
				L-4813
				R-9040

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DANGERS AND UNDERWATER FEATURES (2D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2D130	ROCK			
	Point			
	ACC	Accuracy Category	All required	D-1909
	DAT	Date Category		L-4700
	EXS	Existence Category		L-4702
	HDH	Hydrographic Drying Height		L-4707
	HDI	Hydrographic Depth /Height Information		L-4708
				L-4722
	HDP	Hydrographic Depth		L-4730
	MCP	Material Composition Primary		L-4763
	NAM	Name Category		L-4808
	SOH	Severity of Hazard		L-4872
	VDC	Vertical Datum Category		O-3411
	VDR	Vertical Datum Record		R-2221
	VRC	Vertical Reference Category		R-2222
				R-2806
				R-2916
				R-3707
				R-3708
				R-3709
				T-0836
				L-4709
2D140	SNAG /STUMP			
	Area			
	ACC	Accuracy Category	Area if shown on cartographic source as area	L-4700
	DAT	Date Category		L-4707
	EXS	Existence Category		L-4708
	VRC	Vertical Reference Category		L-4722
				L-4729
				L-4730
				L-4807
				L-4808
				L-4809
				O-3411
				R-2203
				R-2800
				R-2914
				R-3708

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DANGERS AND UNDERWATER FEATURES (2D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2D140	SNAG /STUMP (Continued)			
Point	ACC	Accuracy Category	Point if shown as point on	L-4700
	DAT	Date Category	cartographic source.	L-4702
	EXS	Existence Category		L-4707
	HDI	Hydrographic Depth /Height Information		L-4708
				L-4722
	HDP	Hydrographic Depth		L-4730
	VDC	Vertical Datum Category		L-4808
	VDR	Vertical Datum Record		L-4809
	VRC	Vertical Reference Category		L-4872
				O-3411
				R-2203
				R-2221
				R-2222
				R-2914
				R-3708
				R-3709
2D180	WRECK			
Area	COD	Certainty of Delineation	> 200 m LEN	L-4700
	EPA	Exposed Portion Attribute	OR COD=1	L-4702
	HDI	Hydrographic Depth /Height Information		L-4722
				L-4729
	HDP	Hydrographic Depth		L-4730
	LEN	Length /Diameter		L-4808
	SOH	Severity of Hazard		L-4812
	VDC	Vertical Datum Category		R-2202
	VDR	Vertical Datum Record		R-2221
	VRC	Vertical Reference Category		R-2222
				R-2802
				R-2806
				R-2911
				R-2916
				R-2925
				R-3672
				R-3708

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DANGERS AND UNDERWATER FEATURES (2D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2D180	WRECK (Continued)			
Point	ACC	Accuracy Category	<= 200 m LEN	D-1900
	AOO	Angle of Orientation	OR COD=2	D-1909
	COD	Certainty of Delineation		L-4700
	DAT	Date Category		L-4702
	EPA	Exposed Portion Attribute		L-4707
	EXS	Existence Category		L-4708
	HDI	Hydrographic Depth /Height Information		L-4722
	HDP	Hydrographic Depth		L-4729
	LEN	Length /Diameter		L-4730
	SOH	Severity of Hazard		L-4808
	VDC	Vertical Datum Category		L-4809
	VDR	Vertical Datum Record		L-4872
	VRC	Vertical Reference Category		O-3411
				R-2221
				R-2222
				R-2806
				R-2916
				R-3708
				R-3709
				S-1400
				T-0801
				T-0810

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DEPTH INFORMATION (2E)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2E010	DEPTH CURVE			
Line	ACC	Accuracy Category	(CRV 1, 2, 3, 5, 10, 15, 20, 25,	L-4733
	CRV	Depth Curve or Contour Value	30, 35, 40, 45, 50, 60, 70, 80, 90,	L-4734
	UNI	Units Category	100, 200, 300, 400, 500, 1,000,	L-4776
	VDC	Vertical Datum Category	2,000, 3,000, 4,000, 5,000, 6,000,	O-3408
	VDR	Vertical Datum Record	7,000, 8,000, 9,000, 10,000,	O-3421
			11,000, and 12,000 m)	R-2812
			and UNI 13	R-2813
				R-2814
				R-2827
				R-2828
				R-2869
				R-2873
				R-2874
				R-2875
				R-2876
				R-2882
2E020	SOUNDING			
Point	ACC	Accuracy Category	All required	D-1903
	DAT	Date Category		D-1912
	EXS	Existence Category		D-1913
	HDH	Hydrographic Drying Height		L-4700
	HDP	Hydrographic Depth		L-4702
	SND	Sounding Category		L-4707
	SVC	Sounding Velocity		L-4708
	VDC	Vertical Datum Category		L-4710
	VDR	Vertical Datum Record		L-4711
				L-4722
				O-3403
				O-3405
				O-3406
				O-3411
				R-2200
				R-2207
				R-2222
				R-2224
				R-2807
				R-2864
				R-2908

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: DEPTH INFORMATION (2E)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2E020	SOUNDING (Continued)			
	Point			R-2865
				T-0822
				R-9011
				R-9012
				R-9013
				R-9014
				R-9015
				R-9016
				R-9017
				R-9018
				R-9019
				R-9020
				R-9021
				R-9022
				R-9023
				R-9024
				R-9025
				R-9026
				R-9027
				R-9028
				R-9029
				R-9030
				R-9031
				R-9032
				R-9033
				R-9036

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: BOTTOM FEATURES (2F)

Fcode / FT	Feature Acode	Attribute	Inclusion Condition	Rule
2F010	BOTTOM CHARACTERISTICS			
Point	CSM	Secondary Material Characteristics	All required	L-4701
	MCC	Material Composition Characteristics		L-4706
	MCP	Material Composition Primary		R-2883
	MCS	Material Composition Secondary		R-2890
	MCU	Material Composition Underlying		L-4784
	UMC	Underlying Material Characteristics		R-2892

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PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: TIDE AND CURRENT INFORMATION (2G)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2G010	CURRENT ARROW /FLOW ARROW			
Point	CRN	Current Rate Minimum	(CUR 1 or 2 or 3 or 5)	L-4794
	CRX	Current Rate Maximum	OR CUR 4	R-2436
	CUR	Current Type Category	and When land has a slope of <= 3% SGC (flatland)	R-2891 T-0828
	DOF	Direction of Flow		
	EXS	Existence Category	OR When stream runs through areas of swamp or rice fields.	R-2467
	HS1	Current Information (1)		
	HS2	Current Information (2)		
	SGC	Slope /Gradient Category		
2G040	CURRENT DIAGRAM			
Point	C80	Rate of Current	All required	L-4806 R-2808
	C81	Rate of Current (1)		
	C82	Rate of Current (2)		
	C83	Rate of Current (3)		
	C84	Rate of Current (4)		
	C85	Rate of Current (5)		
	C86	Rate of Current (6)		
	C87	Rate of Current (7)		
	C88	Rate of Current (8)		
	C89	Rate of Current (9)		
	C90	Rate of Current (10)		
	C91	Rate of Current (11)		
	D80	Direction of Current		
	D81	Direction of Current (1)		
	D82	Direction of Current (2)		
	D83	Direction of Current (3)		
	D84	Direction of Current (4)		
	D85	Direction of Current (5)		
	D86	Direction of Current (6)		
	D87	Direction of Current (7)		
	D88	Direction of Current (8)		
	D89	Direction of Current (9)		
	D90	Direction of Current (10)		
	D91	Direction of Current (11)		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: INLAND WATER (2H)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2H010	AQUEDUCT			
Line	ATC	Aqueduct Type Category	(LOC 3 or 4)	L-3970
	EXS	Existence Category	and EXS 28	O-3155
	LEN	Length /Diameter	and >= 75 m LEN	R-2432
	LOC	Location /Origin Category	OR (LOC 1 or 3 or 4)	R-2433
			and (EXS 5 or 6)	
			and >= 75 m LEN	
			OR LOC 1	
			and EXS 28	
			and ATC 2	
			and >= 75 m LEN	
			OR LOC 1	
			and ATC 1	
			and >= 75 m LEN	
Point	ATC	Aqueduct Type Category	ATC = 1,3	R-0034
	EXS	Existence Category	and LOC=3	R-0035
	LOC	Location /Origin Category		
	WID	Width		
2H020	CANAL			
Area	ACC	Accuracy Category	HYC 8	L-4008
	EXS	Existence Category	and (EXS 6 or 32)	L-4702
	HDP	Hydrographic Depth	and >= 25 m WID	R-2316
	HYC	Hydrographic Category	OR HYC 3	R-3673
	LEN	Length /Diameter	and (EXS 5 or 6)	S-1500
	NAM	Name Category	and >= 25 m WID	L-4813
	SLT	Shoreline Type Category		
	WID	Width		
Line	EXS	Existence Category	HYC 8	L-4008
	HDP	Hydrographic Depth	and (EXS 6 or 32)	O-3155
	HYC	Hydrographic Category	and >= 320 m LEN	L-4702
	LEN	Length /Diameter	and < 25 m WID	R-2231
	NAM	Name Category	OR HYC 3	L-4260
	WID	Width	and EXS 5	L-4813
			and >= 320 m LEN	
			and < 25 m WID	

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: INLAND WATER (2H)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2H030	DITCH			
Area	HYC	Hydrographic Category	>= 25 m WID	R-2231
	LEN	Length /Diameter	and HYC 8	R-3673
	WID	Width		
Line	HYC	Hydrographic Category	>= 320 m LEN	R-2231
	LEN	Length /Diameter	and < 25 m WID	R-2267
	WID	Width		O-3155
2H040	FILTRATION /AERATION BEDS			
Area	LMC	Landmark Category	>= 75 m WID	N-0000
	WID	Width	OR LMC 1	
2H050	FISH HATCHERY			
Area	LMC	Landmark Category	>= 75 m WID	R-2231
	WID	Width	OR LMC 1	R-9037
2H060	FLUME			
Line	LEN	Length /Diameter	>= 75 m LEN	R-2231
	LMC	Landmark Category	OR LMC 1	O-3155
	LOC	Location /Origin Category		
2H070	FORD			
Line	LEN	Length /Diameter	>= 25 m LEN	R-2321
			and Associated with road 1P030	
Point	LEN	Length /Diameter	< 25 m LEN	R-2321

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: INLAND WATER (2H)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2H075	**INLAND SHORELINE**			
Line	ACC	Accuracy Category	All required	R-2316
	AHC	Associated Hydrographic Category		R-2426 L-4132
	SLT	Shoreline Type Category		R-2023 R-3735 R-2425 R-2372 D-7010
2H080	**LAKE /POND**			
Area	ARA	Area Coverage Attribute	(HYC 0 or 3 or 6 or 8)	L-3983
	HYC	Hydrographic Category	and >= 2,500 m square ARA	O-3155
	LMC	Landmark Category	OR LMC 1	L-4005
	NAM	Name Category		L-4008
	WID	Width		R-2256
	WSC	Water Salinity Category		R-2270
	ZVL	Z Value		R-2316 R-2425 R-3673
2H090	**LAND SUBJECT TO INUNDATION**			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-3730
	HOC	Hydrographic Origin Category		R-3732 R-3733
2H110	**PENSTOCK**			
Line	LEN	Length /Diameter	>= 75 m LEN	N-0000
	LMC	Landmark Category	OR LMC 1	
	LOC	Location /Origin Category	and < 75 m LEN	
2H120	**RAPIDS**			
Line	LEN	Length /Diameter	>= 25 m WID	R-2429
	WID	Width		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: INLAND WATER (2H)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2H120	RAPIDS (Continued)			
Point	LMC	Landmark Category	< 25 m WID	N-0000
	WID	Width	and LMC 1	
2H130	RESERVOIR			
Area	ARA	Area Coverage Attribute	>= 2,500 m square ARA	L-3971
	EXS	Existence Category	and (EXS = 1 or 5)	O-3155
	LMC	Landmark Category	OR LMC 1	R-2230
	NAM	Name Category		R-9037
	WID	Width		
2H140	RIVER /STREAM			
Area	ACC	Accuracy Category	(HYC 0 or 3 or 6 or 8)	L-4008
	HYC	Hydrographic Category	and >= 25 m WID	R-2316
	LEN	Length /Diameter		R-3673
	NAM	Name Category		S-1500
	SLT	Shoreline Type Category		R-2429
	WID	Width		
Line	EXS	Existence Category	(HYC 0 or 3 or 6 or 8)	L-4008
	HYC	Hydrographic Category	and < 25 m WID	L-4260
	LEN	Length /Diameter	and >= 320 m LEN	R-2316
	LMC	Landmark Category		O-3155
	NAM	Name Category		
	WID	Width		
2H145	RIVER OR STREAM VANISHING POINT			
Point	HFC	Hydrographic Form Category	All required	N-0000
2H150	SALT EVAPORATOR			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-3730
	LMC	Landmark Category	OR LMC 1	R-3732
				R-3733

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: INLAND WATER (2H)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2H160	SABKHA			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	N-0000
2H170	SPRING /WATER-HOLE			
Point	HYC	Hydrographic Category	(HYC 3 or 6 or 8)	O-3155
	SCC	Spring /Well Characteristic Category		R-2231
2H180	WATERFALL			
Line	LEN	Length /Diameter	>= 25 m LEN	L-4008
	NAM	Name Category		L-4260
				L-4813
Point	LEN	Length /Diameter	< 25 m LEN	L-4008
	NAM	Name Category		L-4813

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: MISCELLANEOUS INLAND WATER (2I)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2I010 CISTERN				
Point	LMC	Landmark Category	LMC 001	N-0000
2I020 DAM				
Area	LMC	Landmark Category	>= 25 m WID	L-4008
	MCP	Material Composition Primary		
	NAM	Name Category		
	WID	Width		
Line	LEN	Length /Diameter	>= 25 m LEN	L-4008
	MCP	Material Composition Primary	and < 25 m WID	R-0004
	NAM	Name Category		L-4813
	TUC	Transportation Use Category		
	WID	Width		
Point	LEN	Length /Diameter	< 25 m LEN	L-4008
	LMC	Landmark Category	and LMC 1	O-3155
	MCP	Material Composition Primary		O-3156
	NAM	Name Category		
2I030 LOCK				
Area	LEN	Length /Diameter	>= 25 m WID	R-2371
	NAM	Name Category		R-2232
	WID	Width		L-4008
				L-4813
				R-9037
Point	LMC	Landmark Category	< 25 m WID	R-2232
	WID	Width		R-2371
2I040 SLUICE GATE				
Line	LEN	Length /Diameter	>= 25 m LEN	R-2232
				R-2371
Point	LEN	Length /Diameter	< 25 m LEN	R-2232
	LMC	Landmark Category	and LMC 1	R-2371

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: MISCELLANEOUS INLAND WATER (2I)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2I050	WATER INTAKE TOWER			
Area	LEN	Length /Diameter	> 40 m WID	R-2232 R-3693
	WID	Width		
Point	WID	Width	<= 40 m WID	R-2232

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: SNOW /ICE (2J)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2J020	GLACIAL MORAINE			
Area	WID	Width	>= 320 m WID	R-2316
2J030	GLACIER			
Area	ARA	Area Coverage Attribute	>= 176,400 m square ARA	R-2316 R-3730 R-3732 R-3733 R-9037
2J040	ICE CLIFF			
Line	LEN	Length /Diameter	>= 420 m LEN	R-2399 R-2128
2J060	ICE PEAK, NUNATAK			
Point	HGT	Height Above Surface Level	> 40 m HGT	N-0000
	LMC	Landmark Category	OR LMC 1	
2J065	ICE SHELF			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-2256 R-3730 R-3732 R-3733
2J080	POLAR ICE			
Area	ARA	Area Coverage Attribute	>= 176,400 m square ARA	R-2316 R-3730 R-3732 R-3733 R-9037
	HSA	Hydrographic Seasonal Attribute	and HSA >= 3	

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYDROGRAPHY (2)
SUBCATEGORY: SNOW /ICE (2J)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
2J100	SNOW FIELD /ICE FIELD			
Area	ARA	Area Coverage Attribute	>= 176,400 m square ARA	R-2316
	SIC	Snow /Ice Category		R-3730
				R-3732
				R-3733
				R-9037
2J110	TUNDRA			
Area	ARA	Area Coverage Attribute	>= 176,400 m square ARA	R-2316
	LEN	Length /Diameter		L-4704
	WID	Width		R-3730
				R-3732
				R-3733

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : HYPSOGRAPHY (3)
SUBCATEGORY: RELIEF PORTRAYAL (3A)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
3A010	CONTOUR (LAND)			
Line	HQC	Hypsography Portrayal Category	(HQC 1 or 2 or 3 or 4 or 5 or 6 or 8 or 9 or 14)	L-3998
	MCP	Material Composition Primary		R-2094
	ZVL	Z Value		L-3966
				L-3967
				L-3968
				L-3985
				L-3986
				L-3987
				L-3989
				L-3997
				R-2376
				R-2377
				R-2378
				R-2379
				R-2382
				R-2386
				R-2389
				R-2394
				R-2396
				R-2261
				R-2269
3A030	SPOT ELEVATION			
Point	ACC	Accuracy Category	All required	R-2383
	ELA	Elevation Accuracy		R-2225
	MCP	Material Composition Primary		R-2385
	ZVL	Z Value		L-3984
				L-3802

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : PHYSIOGRAPHY (4)
SUBCATEGORY: EXPOSED SURFACE MATERIAL (4A)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
4A005	ASPHALT LAKE			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-3730 R-3732 R-3733 L-3505
4A010	GROUND SURFACE			
Area	ARA	Area Coverage Attribute	(MCP 6 or 35 or 40 or 43 or 44 or 69)	L-4004 R-2316 R-2392 R-3730 R-3732 R-3733
	MCP	Material Composition Primary	and >= 15,625 m square ARA	
4A020	SALT PAN			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-2316 R-3730 R-3732 R-3733

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : PHYSIOGRAPHY (4)
SUBCATEGORY: LANDFORMS (4B)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
4B010	BLUFF /CLIFF, ESCARPMENT			
Line	HGT	Height Above Surface Level	>= 125 m LEN	R-2387
	LEN	Length /Diameter	and contours coalesce	R-2388
	LMC	Landmark Category	OR >= 100 percent SGC	
	SGC	Slope /Gradient Category	and >= 1.5 m HGT	
			and >= 2,500 m LEN	
4B030	CAVE			
Point	AOO	Angle of Orientation	All required	L-4003
	NAM	Name Category		L-4008
				R-2391
				L-4813
4B060	CREVICE /CREVASSE			
Area	LEN	Length /Diameter	>= 50 m WID	R-3676
	MCP	Material Composition Primary	and >= 420 m LEN	
	WID	Width		
Line	LEN	Length /Diameter	>= 25 m and < 50 m WID	N-0000
	MCP	Material Composition Primary	and >= 420 m LEN	
	WID	Width		
4B071	CUT LINE			
Line	DEP	Depth Below Surface Level	>= 125 m LEN	R-2499
	LEN	Length /Diameter	and >= 3 m DEP	
4B090	EMBANKMENT			
Area	EFI	Embankment /Fill Identifier	EFI 2	R-2202
	HGT	Height Above Surface Level	and >= 50 m WID	R-2802
	LEN	Length /Diameter	and >= 125 m LEN	R-2269
	VRC	Vertical Reference Category	and >= 3 m HGT	
	WID	Width	OR EFI 3	
			and VRC 8	
			and >= 50 m WID	

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : PHYSIOGRAPHY (4)
SUBCATEGORY: LANDFORMS (4B)

Fcode / FT	Feature Acode	Attribute	Inclusion Condition	Rule
4B090	EMBANKMENT	(Continued)		
Line	EFI	Embankment /Fill Identifier	EFI 2	R-2202
	HGT	Height Above Surface Level	and >= 3 m HGT	R-2269
	LEN	Length /Diameter	and >= 125 m LEN	R-2802
	LMC	Landmark Category	and < 50 m WID	
	VRC	Vertical Reference Category	OR EFI 3	
	WID	Width	and VRC 1	
			and >= 75 m LEN	
			OR EFI 3	
			and VRC 8	
			and < 50 m WID	
			OR (EFI 1 or 4)	
			and >= 3 m HGT	
			and >= 125 m LEN	
4B100	ESKER			
Line	LEN	Length /Diameter	>= 75 m LEN	N-0000
	LMC	Landmark Category	OR LMC 1	
4B110	FAULT			
Line	LEN	Length /Diameter	>= 125 m LEN	L-4002
	NAM	Name Category		L-4260
				L-4008
4B115	GEOTHERMAL FEATURE			
Point	GFT	Geothermal Feature Type	All required	T-0303
	LMC	Landmark Category		
4B135	ISLAND			
Area	ARA	Area Coverage Attribute	All required	N-0000

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TABLE I Feature/Attribute category, inclusion conditions and product generation rules.- Continued.

PRODUCT : COMBAT
CATEGORY : PHYSIOGRAPHY (4)
SUBCATEGORY: LANDFORMS (4B)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
4B150	MOUNTAIN PASS			
Point	AOO	Angle of Orientation	All required	L-4008
	NAM	Name Category		O-3154
	ZVL	Z Value		O-3155
				L-4813
4B160	ROCK FORMATION			
Area	HGT	Height Above Surface Level	RKF 1	N-0000
	LMC	Landmark Category	and >= 40 m HGT	
	RKF	Rock Formation Type	OR LMC 1	
			and RKF 1	
Point	HGT	Height Above Surface Level	RKF 3	N-0000
	LMC	Landmark Category	and >= 40 m HGT	
	RKF	Rock Formation Type	OR LMC 1	
			and RKF 3	
4B170	SAND DUNES /SAND HILLS			
Area	ARA	Area Coverage Attribute	>= 176,400 m square ARA	L-3969
	SDO	Sand Dune Orientation		R-2255
	SSC	Structure Shape Category		R-2316
				R-3730
				R-3732
				R-3733
				R-3800
				R-7415
4B180	VOLCANO			
Area	ACC	Accuracy Category	LOC 6	L-4008
	DAT	Date Category		L-4700
	EXS	Existence Category		L-4707
	HGT	Height Above Surface Level		L-4708
	LOC	Location /Origin Category		L-4709
	NAM	Name Category		L-4722
				O-3411

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : PHYSIOGRAPHY (4)
SUBCATEGORY: LANDFORMS (4B)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
4B180	VOLCANO (Continued)			N-0000
	Point	HGT	Height Above Surface Level	LOC 3
		LOC	Location /Origin Category	and <= 40 m HGT

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : VEGETATION (5)
SUBCATEGORY: CROPLAND (5A)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
5A010	CROPLAND (CULTIVATED)			
Area	ARA	Area Coverage Attribute	(FTC 1 or 3)	L-3801
	FTC	Farming Type Category	and (VEG 1 or 4)	R-2316
	LMC	Landmark Category	and >= 15,625 m square ARA	O-3154
	VEG	Vegetation Characteristics	OR LMC 1	R-3730
				R-3732
				R-3733
5A020	HEDGEROW			
Line	LEN	Length /Diameter	>= 500 m LEN	O-3154
	LMC	Landmark Category	OR LMC 1	
5A030	NURSERY			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	L-3801
	LMC	Landmark Category	OR LMC 1	R-2316
				O-3154
				R-3730
				R-3732
				R-3733
5A040	ORCHARD /PLANTATION			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	O-3154
	LMC	Landmark Category	OR LMC 1	R-2316
	PRO	Product Category		R-3730
				R-3732
				R-3733
5A050	VINEYARD /HOPS			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	O-3154
	LMC	Landmark Category	OR LMC 1	R-2316
				R-3730
				R-3732
				R-3733

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : VEGETATION (5)
SUBCATEGORY: RANGELAND (5B)

Fcode / FT	Feature Acode	Attribute	Inclusion Condition	Rule
5B010	GRASSLAND			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	O-3154 R-2316 R-3730 R-3732 R-3733
5B020	SCRUB /BRUSH			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	O-3154 R-2316 R-3730 R-3732 R-3733

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : VEGETATION (5)
SUBCATEGORY: WOODLAND (5C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
5C010	BAMBOO CANE			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	O-3154 R-2316 R-3730 R-3732 R-3733
5C015	FIREBREAK			
Line	LEN	Length /Diameter	>= 1,250 m LEN	R-3694
	WID	Width	and >= 25 m WID	
5C020	OASIS			
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-3730 R-3732 R-3733
Point	ARA	Area Coverage Attribute	< 15,625 m square ARA	O-3154
	VEG	Vegetation Characteristics		O-3155
5C030	TREES			
Area	ARA	Area Coverage Attribute	>= 50% DMT	R-3730
	COC	Conspicuous Object Category	and (TRE 1 or 2 or 3)	R-3732
	COD	Certainty of Delineation	and >= 15,625 m square ARA	R-3733
	DMT	Density Measure (% Tree /Canopy Cover)	OR (VEG 16 or 19) and >= 15,625 m square ARA	L-3801 L-4008
	EXS	Existence Category	OR LMC 1	R-2316
	HGT	Height Above Surface Level	and EXS 42	R-2438
	LMC	Landmark Category	OR DMT >= 25	R-2440
	NAM	Name Category	and < 50%	R-3677
	TRE	Tree Category	and (TRE = 1 or 2 or 3)	
	VEG	Vegetation Characteristics	and ARA >= 15625 m sq.	
	WID	Width		
Point	EXS	Existence Category	EXS 31	O-3154
	LMC	Landmark Category	and LMC 1	

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : VEGETATION (5)
SUBCATEGORY: WETLANDS (5D)

Fcode / FT	Feature Acode	Attribute	Inclusion Condition	Rule
5D010 BOG				
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-2316
	VEG	Vegetation Characteristics		O-3154
				R-3730
				R-3732
				R-3733
5D020 HUMMOCK				
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-2316
				R-3730
				R-3732
				R-3733
5D030 SWAMP				
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-3730
				R-3732
				R-3733
				R-2316
				O-3155
5D040 MARSH				
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	R-3730
				R-3732
				R-3733
				R-2316
				O-3155

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TABLE I Feature/Attribute category, inclusion conditions and product generation rules.- Continued.

PRODUCT : COMBAT
CATEGORY : DEMARCATION (6)
SUBCATEGORY: BOUNDARIES /LIMITS /ZONES (TOPOGRAPHIC) (6A)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
6A000	ADMINISTRATIVE BOUNDARY			
Line	ACC	Accuracy Category	(USE 23 or 26)	D-1655
	NM3	Name 3		L-4873
	NM4	Name 4		R-2358
	USE	Use Status		R-2359
				R-2360
				R-2361
				R-2362
				R-2363
				R-2365
				R-2366
				R-2367
				R-2469
				R-2496
				R-2497
				R-2498
				L-4037
6A020	ARMISTICE LINE			
Line	ACC	Accuracy Category	All required	D-1655
	NM3	Name 3		R-2359
	NM4	Name 4		R-2360
				R-2361
				R-2362
				R-2363
				R-2365
				R-2367
				R-2498
				R-2469
				R-2496
				L-4037

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : DEMARCATION (6)
SUBCATEGORY: BOUNDARIES /LIMITS /ZONES (TOPOGRAPHIC) (6A)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
6A030	CEASE-FIRE LINE			
Line	ACC	Accuracy Category	All required	D-1655 R-2359 R-2360 R-2361 R-2362 R-2363 R-2365 R-2367 R-2498 R-2469 R-2496 L-4037
6A060	DEFACTO BOUNDARY			
Line	ACC	Accuracy Category	(USE 23 or 26)	D-1655 R-2358 R-2359 R-2360 R-2361 R-2362 R-2363 R-2365 R-2367 R-2469 R-2496 R-2498 L-4037
	NM3	Name 3		
	NM4	Name 4		
	USE	Use Status		
6A110	INTERNATIONAL DATE LINE			
Line			All required	L-4817 R-2496

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : DEMARCATION (6)
SUBCATEGORY: BOUNDARIES /LIMITS /ZONES (HYDROGRAPHIC) (6C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
6C040	DREDGED CHANNEL /DREDGED AREA			
Area	ATN	Aids to Navigation	> 50 m WID	L-4702
	DAN	Description of Aids to Navigation		L-4747
	DAT	Date Category		L-4748
	HDP	Hydrographic Depth		L-4827
	MAS	Maintenance Status		R-2205
	VDC	Vertical Datum Category		R-2222
	VDR	Vertical Datum Record		R-2800
	WID	Width		R-2840
				R-2986
Line	ATN	Aids to Navigation	<= 50 m WID	L-4702
	DAN	Description of Aids to Navigation		L-4743
	DAT	Date Category		L-4748
	HDP	Hydrographic Depth		L-4827
	LEN	Length /Diameter		R-2209
	MAS	Maintenance Status		R-2222
	VDC	Vertical Datum Category		R-2800
	VDR	Vertical Datum Record		R-2840
	WID	Width		
6C090	MARITIME LIMIT			
Area	AOO	Angle of Orientation	(MLT 1 or 2 or 3 or 4 or 5 or 11 or	L-4715
	COD	Certainty of Delineation	12 or 13 or 14 or 15 or 16 or 20 or	L-4717
	HOC	Hydrographic Origin Category	21 or 22 or 23)	L-4718
	LEN	Length /Diameter		L-4722
	MLT	Maritime Limit Type		L-4750
	NAM	Name Category		L-4751
	OPS	Operational Status		L-4753
	PRO	Product Category		R-2800
	TXT	Text Attribute		R-2985
	WID	Width		R-3703

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : DEMARCATION (6)
SUBCATEGORY: BOUNDARIES /LIMITS /ZONES (HYDROGRAPHIC) (6C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
6C110	MINE DANGER AREA (HYDRO LIMIT)			
Area	AOO	Angle of Orientation	>= 200 m LEN	L-4715
	COD	Certainty of Delineation	and COD 1	L-4717
	EXS	Existence Category	and EXS 1	L-4718
	LEN	Length /Diameter	OR COD 2	L-4722
	MAS	Maintenance Status	OR EXS 3	L-4753
	WID	Width		L-4756
				O-3413
				R-2800
				R-2809
Point	COD	Certainty of Delineation	MAS 1	L-4722
	EXS	Existence Category	and EXS 1	O-3413
	LEN	Length /Diameter	and < 200 m LEN	R-2809
	MAS	Maintenance Status	and COD 1	
	WID	Width		
6C120	PROHIBITED AREA (HYDRO ONLY)			
Area	AOO	Angle of Orientation	>= 200 m LEN	L-4715
	LEN	Length /Diameter		L-4717
	WID	Width		L-4718
				L-4722
				L-4753
				R-2800
Point	LEN	Length /Diameter	< 200 m LEN	L-4722

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : DEMARCATION (6)
SUBCATEGORY: BOUNDARIES /LIMITS /ZONES (HYDROGRAPHIC) (6C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
6C150	RESTRICTED AREA (HYDRO ONLY)			
Area	AOO	Angle of Orientation	(DTC 16 or 17 or 18 or 19)	L-4715
	DTC	Danger /Obstruction Category	and > 250 m LEN	L-4717
	LEN	Length /Diameter	OR (DTC 12 or 13 or 15 or 20 or 21 or 22)	L-4718
	PRO	Product Category		L-4722
	RAA	Restrictive Area Attribute	and > 250 m WID	L-4753
	USE	Use Status		L-4798
	WID	Width		L-4826
				L-4862
				R-2218
				R-2800
				R-2847
				R-2937
				R-3678
				R-9034
Line	DTC	Danger /Obstruction Category	(DTC 12 or 13 or 15 or 20 or 21 or 22)	L-4743
	LEN	Length /Diameter		L-4862
	PRO	Product Category	and <= 250 m WID	R-2219
	USE	Use Status		R-2220
	WID	Width		R-2847
				R-2937
				R-9034
Point	DTC	Danger /Obstruction Category	(DTC 16 or 17 or 18 or 19)	L-4722
	LEN	Length /Diameter	and <= 250 m LEN	R-3678
	RAA	Restrictive Area Attribute		
	WID	Width		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : DEMARCATION (6)
SUBCATEGORY: BOUNDARIES /LIMITS /ZONES (HYDROGRAPHIC) (6C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
6C165	ROUTE			
Line	ATN	Aids to Navigation	(RTT 2 or 5)	L-4702
	BRR	Bearing and Reciprocal Category		L-4769 R-2209
	BRS	Bearing From Seaward		R-2222
	DAN	Description of Aids to Navigation		R-2820 R-2854
	DOF	Direction of Flow		D-7012
	DRP	Description of Reference Point		L-4813
	EXS	Existence Category		
	HDI	Hydrographic Depth /Height Information		
	HDP	Hydrographic Depth		
	NAM	Name Category		
	RTT	Route Type Attribute		
	VDC	Vertical Datum Category		
	VDR	Vertical Datum Record		
6C170	SAFETY FAIRWAY			
Area	WID	Width	> 50 m WID	L-4747 L-4772 R-2986
Line	WID	Width	<= 50 m WID	L-4743
6C177	SWEPT AREA			
Area	DAT	Date Category	All required	R-2984
	HDP	Hydrographic Depth		R-2822
	VDC	Vertical Datum Category		R-2222
	VDR	Vertical Datum Record		L-4702 L-4771

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : DEMARCATION (6)
SUBCATEGORY: BOUNDARIES /LIMITS /ZONES (HYDROGRAPHIC) (6C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
6C210	WORK IN PROGRESS AREA			
Area	AOO	Angle of Orientation	> 50 m WID	R-2857
	ATN	Aids to Navigation	and WPC 2	L-4706
	COD	Certainty of Delineation	OR WPC 1	L-4753
	DAN	Description of Aids to Navigation	OR COD 2	L-4774 L-4722
	DAT	Date Category		
	LEN	Length /Diameter		
	WID	Width		
	WPC	Work in Progress Category		
Line	COD	Certainty of Delineation	COD 1	L-4706
	DAT	Date Category	and <= 50 m WID	L-4774
	LEN	Length /Diameter	and WPC 2	R-2857
	WID	Width		
	WPC	Work in Progress Category		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : GENERAL (9)
SUBCATEGORY: CONTROL POINTS (9B)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
9B030	BOUNDARY MARKER			
	Point NAM	Name Category	All required	R-2357
9B035	CONTROL POINT			
	Point CPA	Control Point Attribute	(CPA 1 or 2 or 3 or 5)	R-2374
	ZVL	z Value		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : GENERAL (9)
SUBCATEGORY: MAGNETIC VARIATION INFO (9C)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
9C040	MAGNETIC DISTURBANCE AREA			
	Area		All required	L-4705 L-4737 L-4722
	VAV	Variation Anomaly Value		

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TABLE I. Feature/Attribute category, inclusion conditions and product generation rules - Continued.

PRODUCT : COMBAT
CATEGORY : GENERAL (9)
SUBCATEGORY: MISCELLANEOUS (9D)

Fcode FT	Feature Acode	Attribute	Inclusion Condition	Rule
9D015	POINT OF CHANGE			
Point	PCI	Point of Change Identifier	All required	R-2173 R-2176 R-2175 L-3958 R-2357 R-2430 R-2498
9D020	VOID COLLECTION AREA			N-0000
Area	ARA	Area Coverage Attribute	>= 15,625 m square ARA	
	VCA	Void Collection Attribute	and (VCA 2 or 3 or 6 or 7)	
	VCT	Void Collection Type		

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APPENDIX A

COMBAT CHART STYLE SHEET (EAST-WEST)

10. SCOPE

10.1 Scope. This Appendix is a graphic illustration of the design, composition, and location of the margin data. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards and handbooks. This section is not applicable to this Appendix.

20.2.1 Other government documents, drawings, and publications. Technical manual 8358.1. Copies of DMA TM 8358.1 are available to Department of Defense users, from the Defense Mapping Agency Combat Support Center, ATTN; DDCP, Washington, D.C. 20315-0020. All other request should be directed to the National Technical Information Center, Cameron Station, Alexandria, VA. 22315-6145

20.2 Non-government publications. This section is not applicable to this Appendix.

30. COMBAT CHART STYLE SHEET

30.1 Style sheet. See next page for style sheet information.

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APPENDIX B

COMBAT CHART STYLE SHEET (NORTH-SOUTH)

10. SCOPE

10.1 Scope. This Appendix is a graphic illustration of the design, composition, and location of the margin data. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards and handbooks. This section is not applicable to this Appendix.

20.2.1 Other government documents, drawings, and publications. Technical manual 8358.1. Copies of DMA TM 8358.1 are available to Department of Defense users, from the Defense Mapping Agency Combat Support Center, ATTN; DDCP, Washington, D.C. 20315-0020. All other request should be directed to the National Technical Information Center, Cameron Station, Alexandria, VA. 22315-6145

20.2 Non-government publications. This section is not applicable to this Appendix.

30. COMBAT CHART STYLE SHEET

30.1 Style sheet. See next page for style sheet information.

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CONCLUDING MATERIAL

Custodians:
DMA - MP

Preparing activity:
DMA - MP

Review activities:
Navy - NO
Army - PO
Marine Corps - MC

(project MCGT-0023)

User activities:
Army: USA-FSTC-CB1; HQ USEUCOM-ECJ2-T; OIC-MI BN; USSOCOM-SCJ2-P.
DIA: DIA-DB.
DMA: DMAAC-DAP; DMAAC-PRA; DMACSC EUR DET; DMACSC LANT-SP-238; DMACSC LAT AM; DMAHTC LUO-LUM; DMAHTC LUO-VRM; DMAHTC SXO-SXM.
DMS: DMS-MTM.
Navy: CINCLANT-62; CINCPAC-02M2; CINCUSNAEUR-N39; HQ FMFEUR-G2; HQ WESTCOM-APIN-MCG.
NSA: NSA-T5141.
USGS: USGS-C/NMD.

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